

Application of Quantum Error Correction and Fault-Tolerance in Quantum Metrology

Sisi Zhou

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Outline

- Part 1: Introduction to Quantum Metrology
- Part 2: Heisenberg Limit vs Standard Quantum Limit
(— — Coffee Break? — —)
- Part 3: Introduction to Quantum Error Correction
- Part 4: Quantum Error Correction for Quantum Metrology
 - Case Study I: Phase estimation under bit-flip noise
 - Case Study II: ZZZ interaction estimation under single-qubit noise
- Part 5: Fault Tolerance in Quantum Metrology (An Example)

Part 1

INTRODUCTION TO QUANTUM METROLOGY

* CLASSICAL PARAMETER ESTIMATION

* CRAMÉR-RAO BOUND

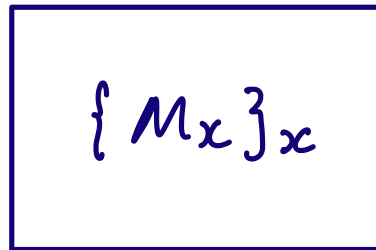
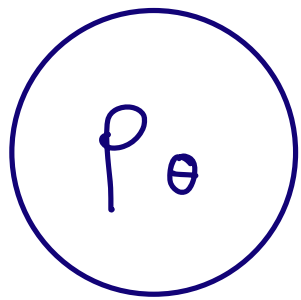
* QUANTUM PARAMETER ESTIMATION

QUANTUM METROLOGY

Quantum State

Quantum Measurement

Probability Distribution



$$p(x|\theta)$$

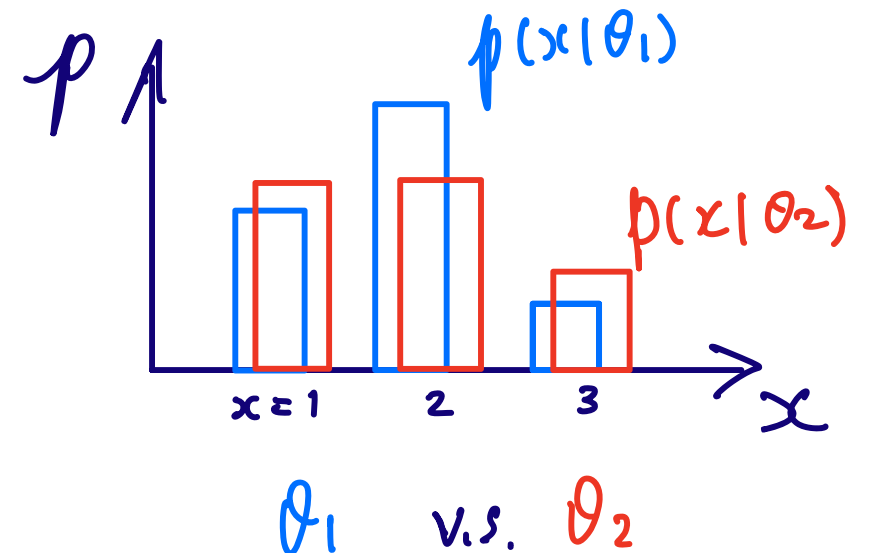
Density matrix

$$\rho_\theta \geq 0, \quad \text{Tr}(\rho_\theta) = 1,$$

POVM

$$M_x \geq 0, \quad \sum_x M_x = \mathbb{I}$$

$$p(x|\theta) = \text{Tr}(\rho_\theta M_x)$$



MEAN SQUARE ERROR

* Estimator $\hat{\theta}(x) : \underline{X} \longrightarrow \Theta$
(Domain of x) (Domain of θ)

$$\hat{\theta}(x_1, \dots, x_n) : \underline{X}^n \longrightarrow \Theta$$

* Unbiased Estimator

$$\mathbb{E}_x(\hat{\theta}) = \theta \quad \text{--- True Value of } \theta$$

* Mean Square Error (MSE)

$$V_{\theta} = \mathbb{E}_x[(\hat{\theta} - \theta)^2] \quad \text{--- Characterizes estimation precision}$$

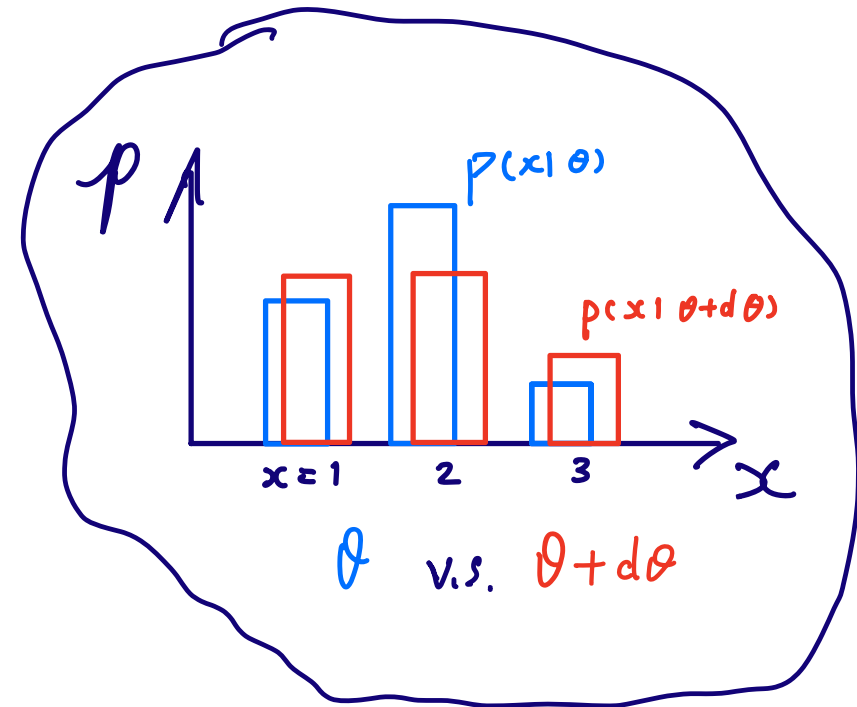
CRAMÉR-RAO BOUND

* Lower bound on MSE $V_\theta \geq \frac{1}{N \cdot F}$

N : number of samples

F : (classical) Fisher information FI

$$F = \sum_x \frac{1}{p(x|\theta)} \left[\frac{\partial p(x|\theta)}{\partial \theta} \right]^2$$



* FI can be interpreted as the second-order derivative of FIDELITY

$$\text{Fidelity: } f(p_\theta, p_{\theta+d\theta}) = \sum_x \sqrt{p(x|\theta) p(x|\theta+d\theta)}$$

$$\text{Taylor Expansion: } = 1 - \frac{(d\theta)^2}{8} \sum_x \frac{(\partial_\theta p(x|\theta))^2}{p(x|\theta)} + O(d\theta^3) = 1 - \frac{(d\theta)^2}{8} \cdot F + O(d\theta^3)$$

PROOF OF CRAMÉR-RAO BOUND

* Unbiased Estimator

$$\mathbb{E}_x(\hat{\theta}) = \theta \quad \text{--- True Value of } \theta$$

$$* \begin{cases} \sum_x p(x|\theta) \hat{\theta}(x) = \theta \\ \sum_x p(x|\theta) = 1 \end{cases} \xrightarrow{\text{Derivative}} \begin{cases} \sum_x \partial_{\theta} p(x|\theta) \hat{\theta}(x) = 1 \\ \sum_x \partial_{\theta} p(x|\theta) = 0 \end{cases}$$

$$\Rightarrow \sum_x \boxed{\frac{\partial_{\theta} p(x|\theta)}{\sqrt{p(x|\theta)}}} \cdot \boxed{\sqrt{p(x|\theta)} (\hat{\theta}(x) - \theta)} = 1$$

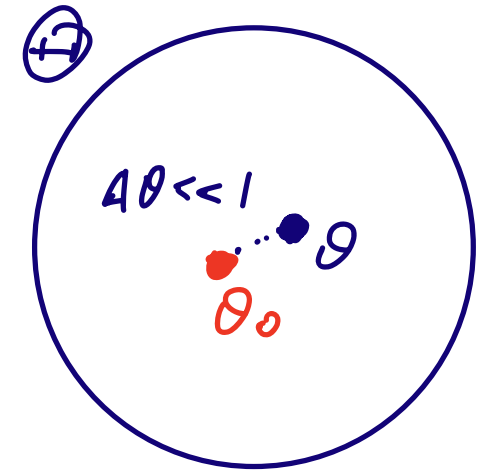
$$\xrightarrow{\text{Cauchy-Schwarz}} \sum_x \frac{(\partial_{\theta} p(x|\theta))^2}{p(x|\theta)} \times \sum_x p(x|\theta) (\hat{\theta}(x) - \theta)^2 \geq 1$$

$$FI \times V \geq 1$$

ATTAINABILITY OF CR BOUND

* Locally unbiased estimator at $\theta = \theta_0$:

$$\hat{\theta}_{l.u.}(x; \theta_0) = \theta_0 + \frac{1}{F(\theta_0)} \cdot \frac{\partial_\theta p(x|\theta_0)|_{\theta=\theta_0}}{p(x|\theta_0)}$$



Satisfying $E_x(\hat{\theta}_{l.u.})|_{\theta=\theta_0} = \theta_0$, and $V(\hat{\theta}_{l.u.})|_{\theta=\theta_0} = \frac{1}{F(\theta_0)}$

$$\frac{\partial}{\partial \theta} E_x(\hat{\theta}_{l.u.})|_{\theta=\theta_0} = 1. \quad (\text{CR bound})$$

(locally unbiasedness condition)

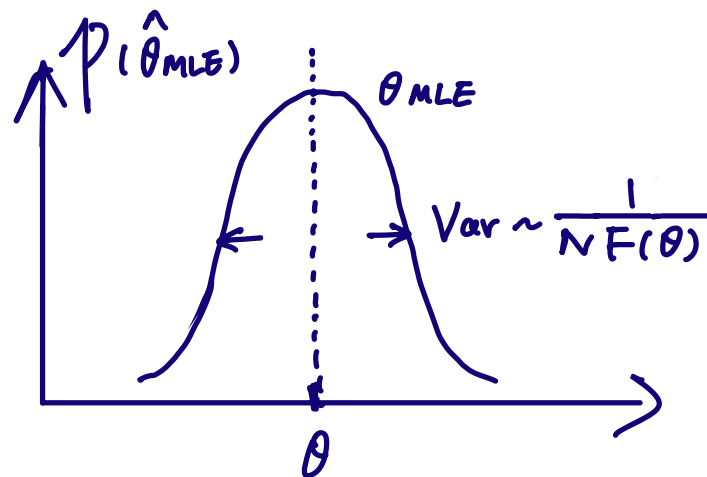
ATTAINABILITY OF CR BOUND

* Maximally likelihood estimation

$$\hat{\theta}_{MLE} = \arg \max_{\theta} p(x_1, x_2, \dots, x_N | \theta)$$

* Under certain regularity conditions,

$$\sqrt{N} (\hat{\theta}_{MLE} - \theta) \xrightarrow{d} \text{Normal}(0, F(\theta)^{-1})$$



QUANTUM FISHER INFORMATION

* Classical FI

$$F(\{p(x|\theta)\}) = \sum_x \frac{1}{p(x|\theta)} \left[\frac{\partial p(x|\theta)}{\partial \theta} \right]^2, \quad p(x|\theta) = \text{Tr}(\rho_\theta M_x)$$

$$= \mathbb{E}_x \left[\left(\frac{\partial}{\partial \theta} \log p(x|\theta) \right)^2 \right]$$


Expectation

 logarithmic derivative

* Quantum FI (QFI)

$$F(\rho_\theta) = \max_{\{M_x\}_x} F(p(x|\theta)) = \text{Tr}(\rho_\theta L_\theta^2)$$

Symmetric SLD
logarithmic
derivative


Expectation

QUANTUM FISHER INFORMATION

$$* \quad \boxed{\partial_\theta \log p(x|\theta)} = \frac{\partial_\theta p(x|\theta)}{p(x|\theta)}$$

$\Downarrow$

$$\frac{1}{2} (\underbrace{L_\theta \rho_\theta}_{\text{SLD}} + \rho_\theta \underbrace{L_\theta}_{\text{Hermitian}}) = \partial_\theta \rho_\theta$$

* QFI can be interpreted as the second-order derivative of FIDELITY

$$\text{Fidelity: } f(\rho_\theta, \rho_{\theta+d\theta}) = \text{Tr}(\sqrt{\sqrt{\rho_\theta} \rho_{\theta+d\theta} \sqrt{\rho_\theta}})$$

$$\text{Taylor Expansion:} \quad = 1 - \frac{(d\theta)^2}{8} \underbrace{F(\rho_\theta)}_{\text{QFI}} + O(d\theta^3)$$

PROPERTIES OF QFI (FI)

$$* \quad \underset{\text{FI}}{F}(\{p(x|\theta)\}) = \underset{\text{QFI}}{F}(\rho_\theta), \quad \rho_\theta = \sum_x p(x|\theta) |x\rangle\langle x|$$

$$* \text{ Additivity: } F(\rho_\theta \otimes \sigma_\theta) = F(\rho_\theta) + F(\sigma_\theta)$$

$$\Rightarrow F(\rho_\theta^{\otimes n}) = n F(\rho_\theta)$$

$$* \text{ Monotonicity: } F(\underbrace{E(\rho_\theta)}_{\text{any parameter-independent quantum channel}}) \leq F(\rho_\theta)$$

any parameter-independent quantum channel

$$* \text{ Convexity: } F(p_1 \rho_\theta + p_2 \sigma_\theta)$$

$$\leq p_1 F(\rho_\theta) + p_2 F(\sigma_\theta)$$

QUANTUM FISHER INFORMATION

* Pure state QFI Calculation

$$\rho_\theta = |\psi_\theta\rangle\langle\psi_\theta|, \quad L_\theta = 2[|\partial_\theta\psi_\theta\rangle\langle\psi_\theta| + |\psi_\theta\rangle\langle\partial_\theta\psi_\theta|]$$

Verify SLD :

$$\begin{aligned} \frac{1}{2}[L_\theta\rho_\theta + \rho_\theta L_\theta] &= |\partial_\theta\psi_\theta\rangle\langle\psi_\theta| + |\psi_\theta\rangle\langle\partial_\theta\psi_\theta| \\ &+ |\psi_\theta\rangle\langle\psi_\theta| \left(\underbrace{\langle\psi_\theta|\partial_\theta\psi_\theta\rangle + \langle\partial_\theta\psi_\theta|\psi_\theta\rangle}_{= \partial_\theta(\langle\psi_\theta|\psi_\theta\rangle) = \partial_\theta 1 = 0} \right) \\ &= |\partial_\theta\psi_\theta\rangle\langle\psi_\theta| + |\psi_\theta\rangle\langle\partial_\theta\psi_\theta| = \partial_\theta(|\psi\rangle\langle\psi|) \end{aligned}$$

$$\begin{aligned} F(|\psi_\theta\rangle) &= \text{Tr}(\rho_\theta L_\theta^2) = 4\langle\psi| \left[|\partial_\theta\psi\rangle\langle\psi|\partial_\theta\psi\rangle\langle\psi| + |\partial_\theta\psi\rangle\langle\psi|\psi\rangle\langle\partial_\theta\psi| + \right. \\ &\quad \left. |\psi\rangle\langle\partial_\theta\psi|\psi\rangle\langle\partial_\theta\psi| + |\psi\rangle\langle\partial_\theta\psi|\partial_\theta\psi\rangle\langle\psi| \right] |\psi\rangle \\ &= 4 \left[\langle\psi|\partial_\theta\psi\rangle\langle\psi|\partial_\theta\psi\rangle + \langle\psi|\partial_\theta\psi\rangle\langle\partial_\theta\psi|\psi\rangle + \langle\partial_\theta\psi|\psi\rangle\langle\partial_\theta\psi|\psi\rangle + \langle\partial_\theta\psi|\partial_\theta\psi\rangle \right] \\ &= 4 \left[\langle\partial_\theta\psi|\partial_\theta\psi\rangle - \langle\psi|\partial_\theta\psi\rangle\langle\partial_\theta\psi|\psi\rangle \right] \end{aligned}$$

QUANTUM FISHER INFORMATION

* Mixed state QFI calculation

$$\rho_\theta = \sum_i \lambda_i |\psi_i\rangle\langle\psi_i|$$


functions of θ

$$\begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_d \end{bmatrix}$$

$$\langle\psi_k|L_i|\psi_l\rangle = \begin{cases} \frac{2\langle\psi_k|\partial_\theta\rho_\theta|\psi_l\rangle}{\lambda_k + \lambda_l}, & \text{when } \lambda_k + \lambda_l > 0 \\ 0, & \text{otherwise} \end{cases}$$

$$F(\rho_\theta) = \text{Tr}(\rho_\theta L_\theta^2) = 2 \sum_{\lambda_k + \lambda_l > 0} \frac{|\langle\psi_k|\partial_\theta\rho_\theta|\psi_l\rangle|^2}{\lambda_k + \lambda_l}$$

SUMMARY OF PART 1

- * Setting of quantum metrology
- * CR Bound
- * Fisher Information
- * Quantum Fisher Information

PART 2

HEISENBERG LIMIT

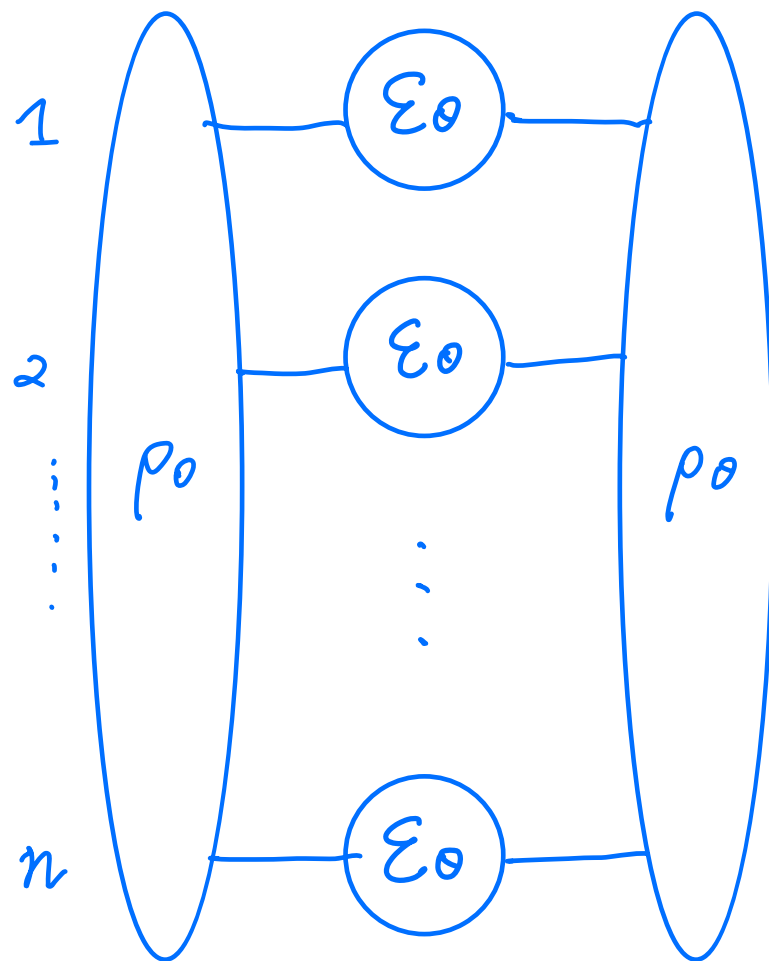
VS STANDARD QUANTUM LIMIT

* QUANTUM CHANNEL ESTIMATION

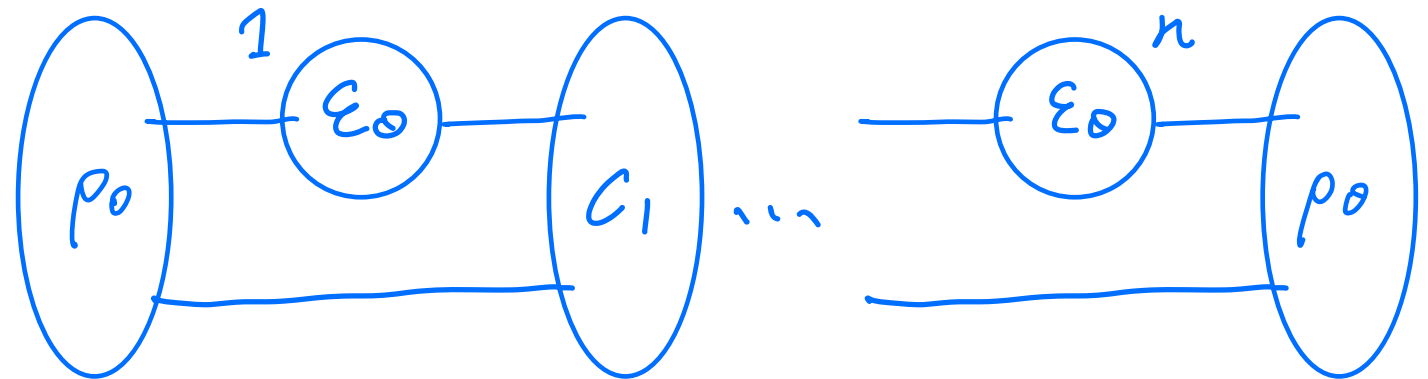
* NOISELESS SCENARIO : PHASE ESTIMATION

* NOISY SCENARIO : PHASE ESTIMATION

QUANTUM CHANNEL ESTIMATION



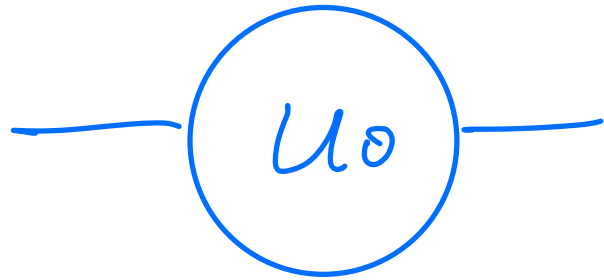
Parallel



Sequential

- * Heisenberg Limit : $F(\rho_0) \sim n^2$
HL, optimal
- * Standard Quantum Limit : $F(\rho_0) \sim n$
SQL, achievable with product state

PHASE ESTIMATION



$$U_0 = \exp(-i Z \theta), \quad Z = \text{Pauli } Z$$

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

* Single-qubit case

$$|+\rangle := \frac{|0\rangle + |1\rangle}{\sqrt{2}} \xrightarrow{U_0} \frac{e^{-i\theta}|0\rangle + e^{i\theta}|1\rangle}{\sqrt{2}} \quad \therefore \psi_\theta$$

$$\{M_\pm\} = \left\{ \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}, \frac{|0\rangle \pm i|1\rangle}{\sqrt{2}} \right\}$$

$$p(\pm|\theta) = \text{Tr}(M_\pm \psi_\theta) = \cos^2 \theta \quad \text{or} \quad \sin^2 \theta$$

PHASE ESTIMATION

* FI Calculation:

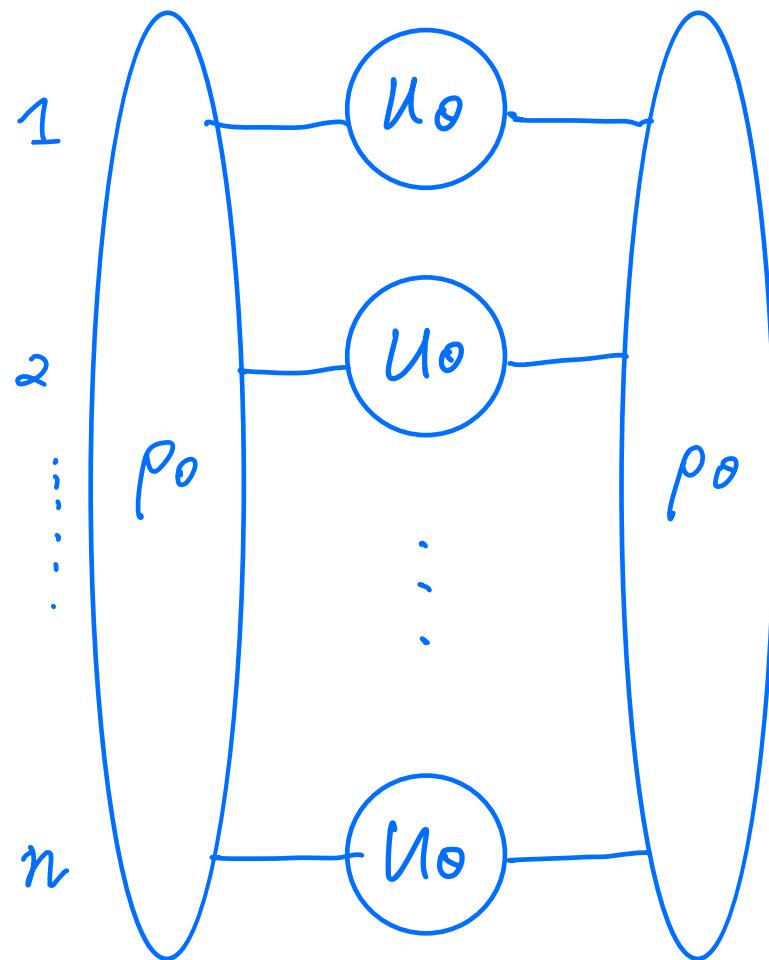
$$F(p(\pm 10)) = \frac{(\partial_0 \cos^2 \theta)^2}{\cos^2 \theta} + \frac{(\partial_0 \sin^2 \theta)^2}{\sin^2 \theta} = 4(\sin^2 \theta + \cos^2 \theta) = 4$$

* QFI calculation

$$\begin{aligned} F(40) &= 4 [\langle 204 | 204 \rangle - \langle 4 | 204 \rangle \langle 204 | 4 \rangle] \\ &= 4(1 - 0) = 4 \end{aligned}$$

* $\{M_{\pm}\}$ is optimal

PHASE ESTIMATION ——— PARALLEL



* n -qubit GHZ state

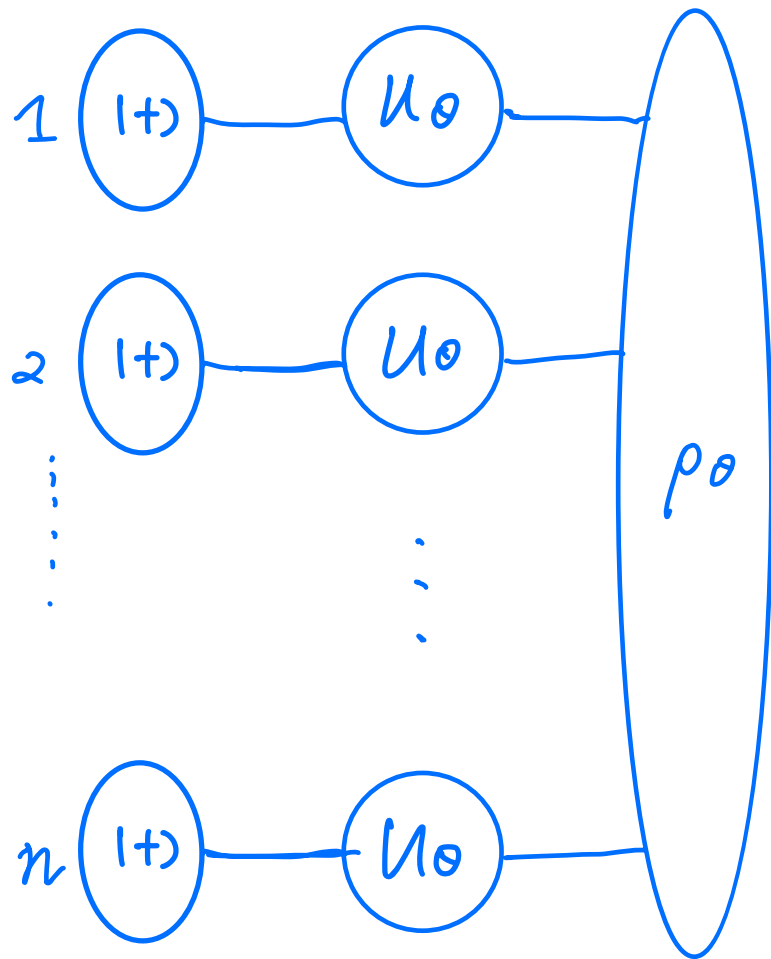
(long-range entanglement)

$$|\psi_0\rangle = \frac{1}{\sqrt{2}} (|0\dots 0\rangle + |1\dots 1\rangle) \xrightarrow[U_0]{\otimes n}$$

$$|\psi_\theta\rangle = \frac{1}{\sqrt{2}} (e^{-i\theta \cdot n} |0\dots 0\rangle + e^{i\theta \cdot n} |1\dots 1\rangle)$$

$$QFI(\psi_\theta) = 4n^2 \sim ML$$

PHASE ESTIMATION ——— PARALLEL



* n -qubit product state

$$|\psi_0\rangle = |+\rangle^{\otimes n} \xrightarrow{U_\theta^{\otimes n}} \left[\frac{e^{-i\theta}|0\rangle + e^{i\theta}|1\rangle}{\sqrt{2}} \right]^{\otimes n}$$

Using additivity of QFI,

$$\text{QFI}(\psi_\theta) = 4n \sim \text{SQL}$$

PHASE ESTIMATION — SEQUENTIAL



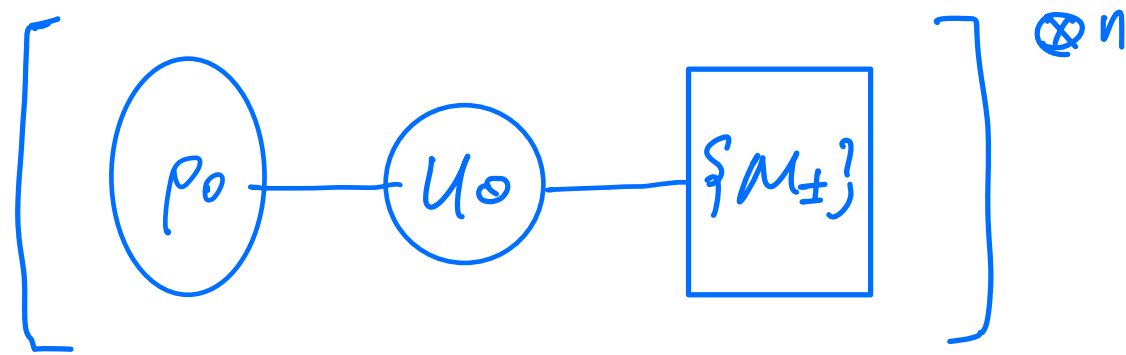
*: Input state: $|+\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$

*: Output state: $|\psi_\theta\rangle = [e^{-in\theta} |0\rangle + e^{in\theta} |1\rangle] / \sqrt{2}$

$$F(\psi_\theta) = 4n^2 \sim HL$$

No control needed.

PHASE ESTIMATION — SEQUENTIAL



measure n times

*: Input state: $|+\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$

*: Output state: $|\psi_0\rangle = \left[e^{-i\theta} |0\rangle + e^{i\theta} |1\rangle \right] / \sqrt{2}$

*: Repeat n times

Total FI: $n \times F(\psi_0) = 4n \sim \text{SQL}$

UNITARY ESTIMATION

* $U_\theta = \exp(-iH\theta)$, H : Hermitian operator
(Hamiltonian)

* Parallel:

optimal input state: $\frac{1}{\sqrt{2}} \left[|\lambda_{\min}^{\otimes n}\rangle + |\lambda_{\max}^{\otimes n}\rangle \right]$

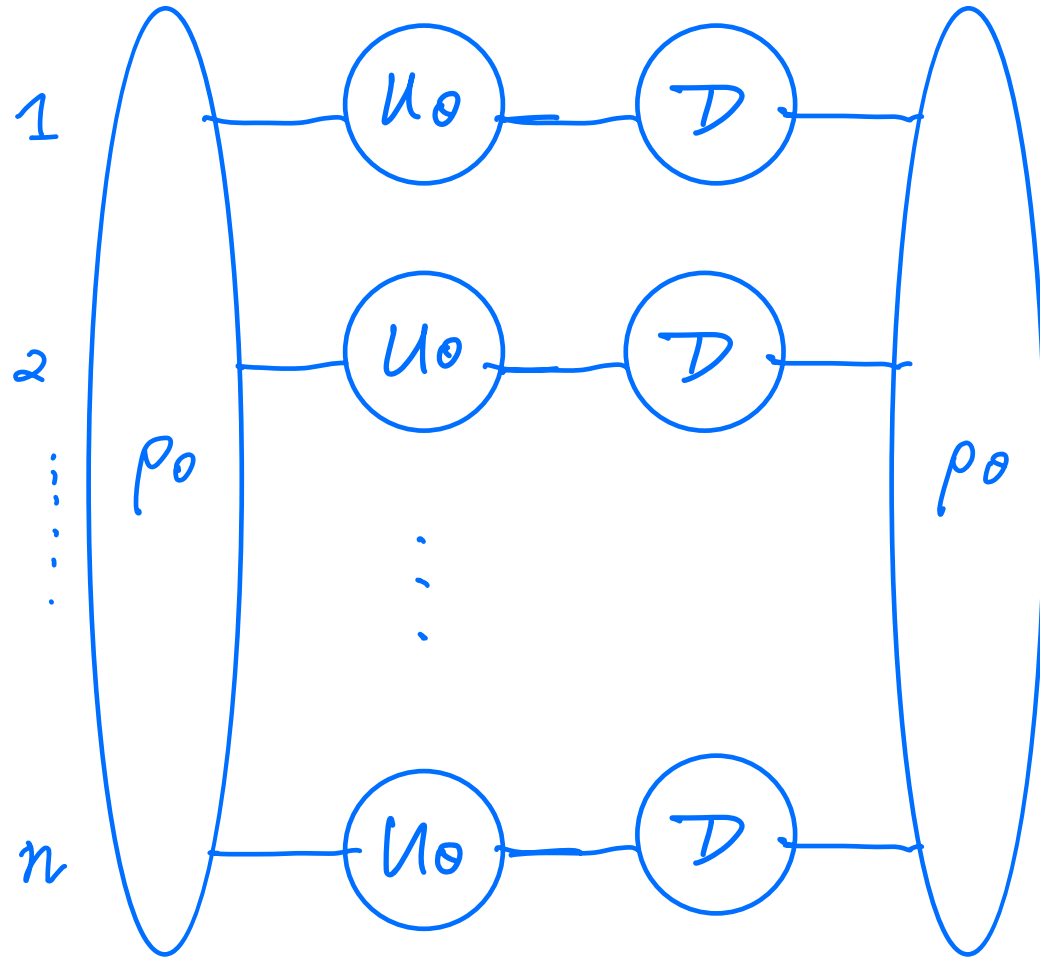
$$QFI = (\lambda_{\max} - \lambda_{\min})^2 n^2 \sim HL \rightarrow \text{OPTIMAL}$$

* Sequential:

optimal input state: $\frac{1}{\sqrt{2}} \left[|\lambda_{\min}\rangle + |\lambda_{\max}\rangle \right]$, some QFI

* To achieve HL: long-range entanglement, or long-time coherence

NOISY PHASE ESTIMATION



D : noise channel

$$\text{Pauli } X: \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$\uparrow$

* Bit-flip noise: $(1-p)(\cdot) + p X(\cdot)X$

* Dephasing noise: $(1-p)(\cdot) + p Z(\cdot)Z$

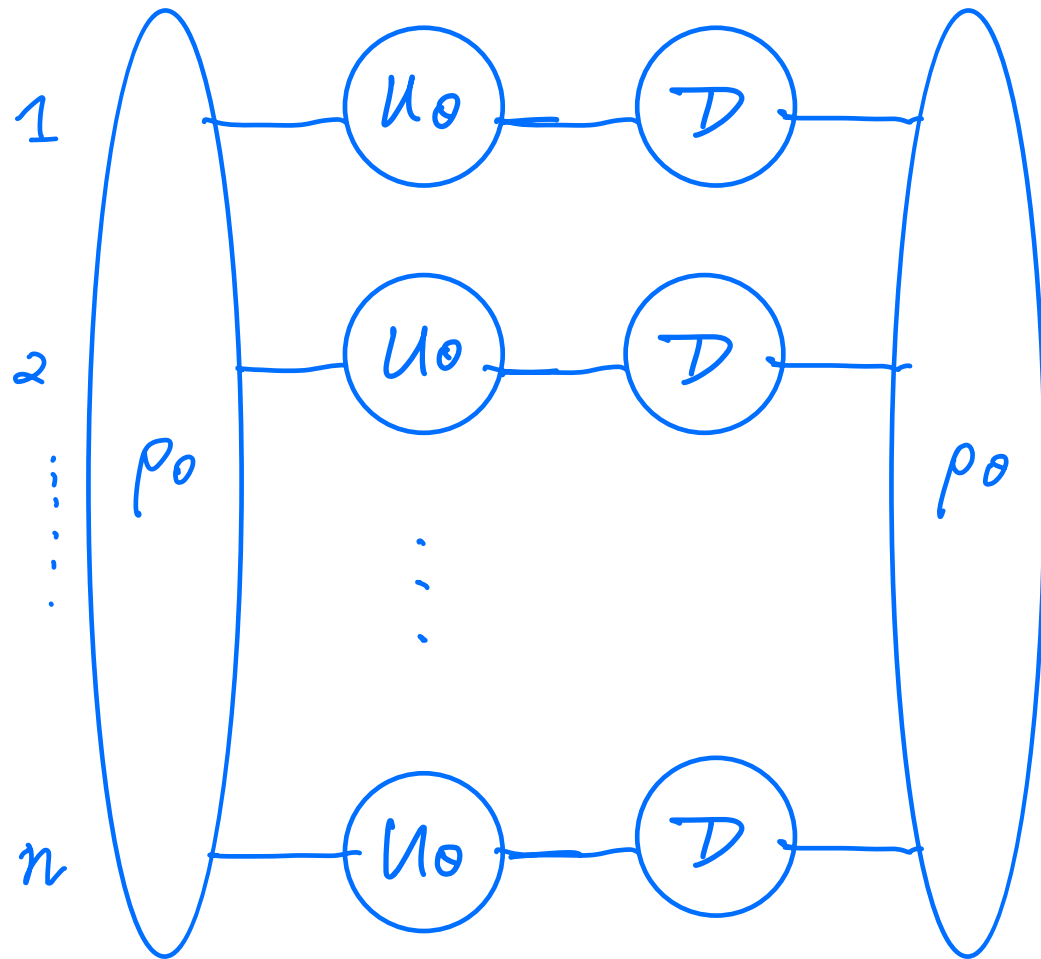
* Depolarizing noise: $(1-p)(\cdot) + p (\mathbb{I}/2)$

||

$$(1 - \frac{3}{4}p)(\cdot) + \frac{p}{4} X(\cdot)X + \frac{p}{4} Y(\cdot)Y + \frac{p}{4} Z(\cdot)Z$$

(mixture of x, y, z)

NOISY PHASE ESTIMATION ——— PARALLEL



* slight decrease in QFI
if \$D\$ acts before \$U_\theta\$

* \$n\$-qubit GHZ state + Bit-flip noise

$$\frac{1}{\sqrt{2}} (|0^{\otimes n}\rangle + |1^{\otimes n}\rangle) \xrightarrow{U_\theta^{\otimes n}} \frac{1}{\sqrt{2}} (e^{-in\theta} |0^{\otimes n}\rangle + e^{in\theta} |1^{\otimes n}\rangle)$$

$$\{M_\pm\} = \left\{ \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}, \frac{\langle 0| \pm \langle 1|}{\sqrt{2}} \right\}$$

Measure parity: (even if product is +1)

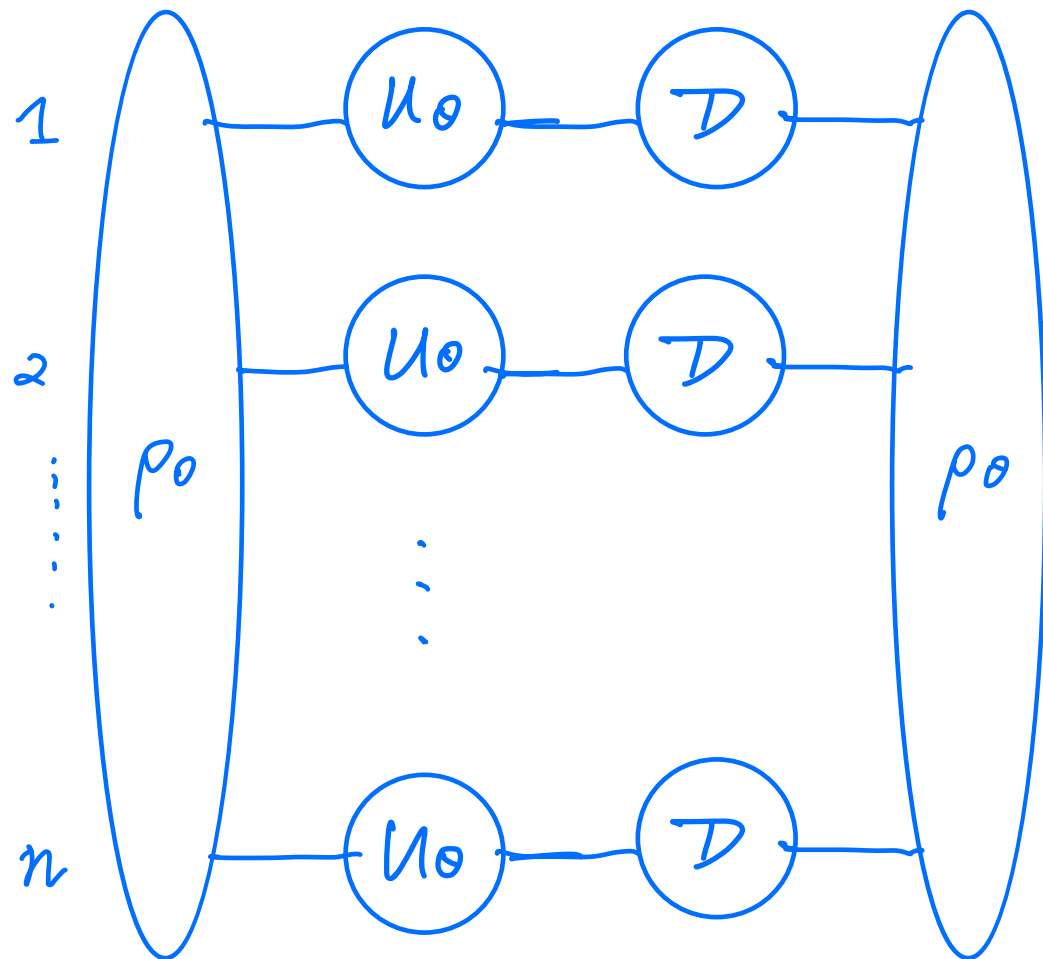
$$p(\text{even}) = \cos^2 n\theta$$

$$p(\text{odd}) = \sin^2 n\theta$$

* \$\{M_\pm\}\$ outcome invariant under bit-flip

$$\Rightarrow QFI = 4n^2 \quad \sim \text{the same as noiseless case}$$

NOISY PHASE ESTIMATION ——— PARALLEL



* \$n\$-qubit GHZ state + **Dephasing** noise

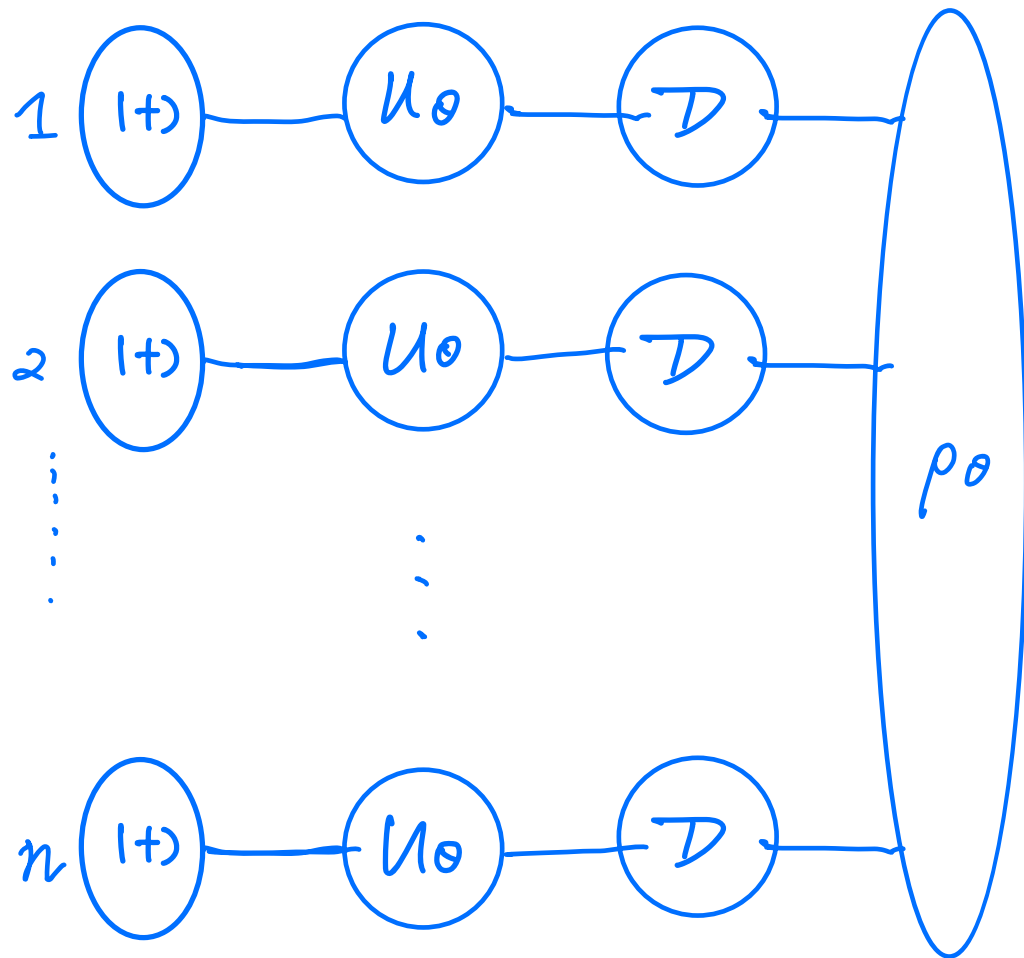
$$\frac{1}{\sqrt{2}} (|0^{\otimes n}\rangle + |1^{\otimes n}\rangle) \xrightarrow{U_\theta^{\otimes n}} \frac{1}{\sqrt{2}} (e^{-in\theta} |0^{\otimes n}\rangle + e^{in\theta} |1^{\otimes n}\rangle)$$

$$\xrightarrow{D^{\otimes n}} \frac{1}{2} \begin{bmatrix} 1 & (1-2p)^n e^{-i2n\theta} \\ (1-2p)^n e^{i2n\theta} & 1 \end{bmatrix}$$

$$|\vec{0}\rangle + |\vec{1}\rangle \xrightarrow{Z^S = Z^{s_1} \otimes Z^{s_2} \otimes \dots \otimes Z^{s_n}} |\vec{0}\rangle + (-1)^{|S|} |\vec{1}\rangle \quad \text{two-level}$$

$$QFI = 4n^2 (1-2p)^{2n} \quad \text{exponential decay}$$

NOISY PHASE ESTIMATION ——— PARALLEL

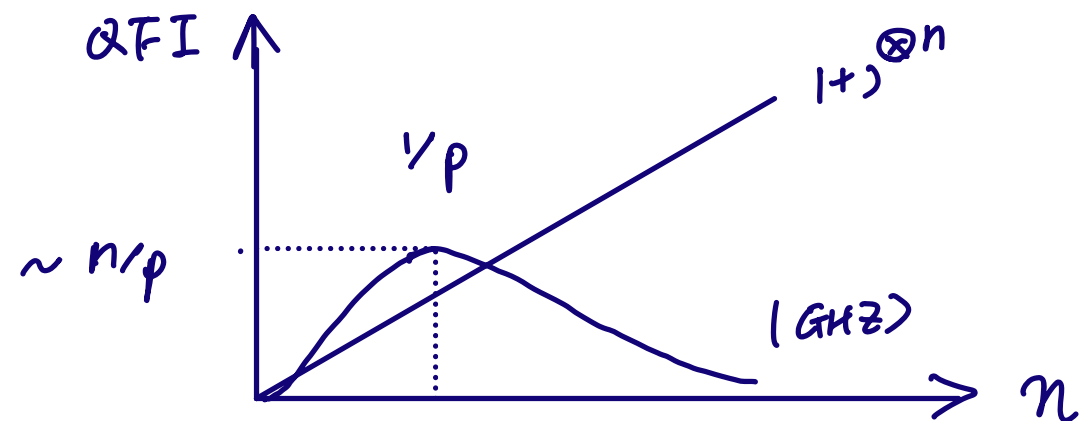


* n -qubit **product state** + **Dephasing** noise

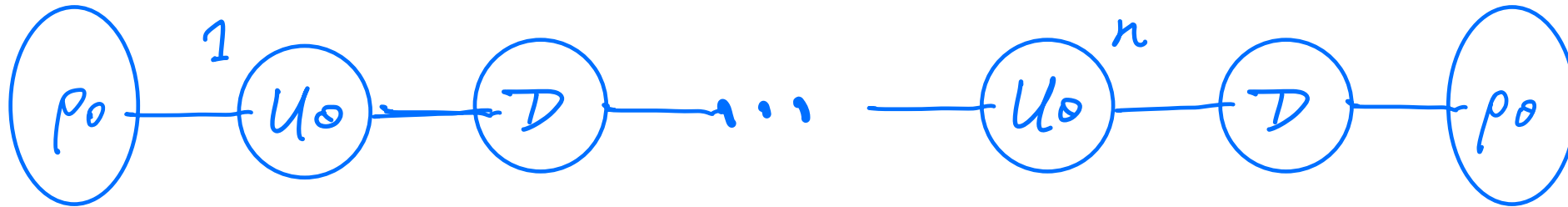
$$\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \xrightarrow{U_\theta} \frac{1}{\sqrt{2}} (e^{-i\theta} |0\rangle + e^{i\theta} |1\rangle)$$

$$\xrightarrow{D} \frac{1}{2} \begin{bmatrix} 1 & (1-2p) e^{-i2\theta} \\ (1-2p) e^{i2\theta} & 1 \end{bmatrix}$$

$$QFI = 4n (1-2p)^2 \sim \text{SQL}$$



NOISY PHASE ESTIMATION — SEQUENTIAL



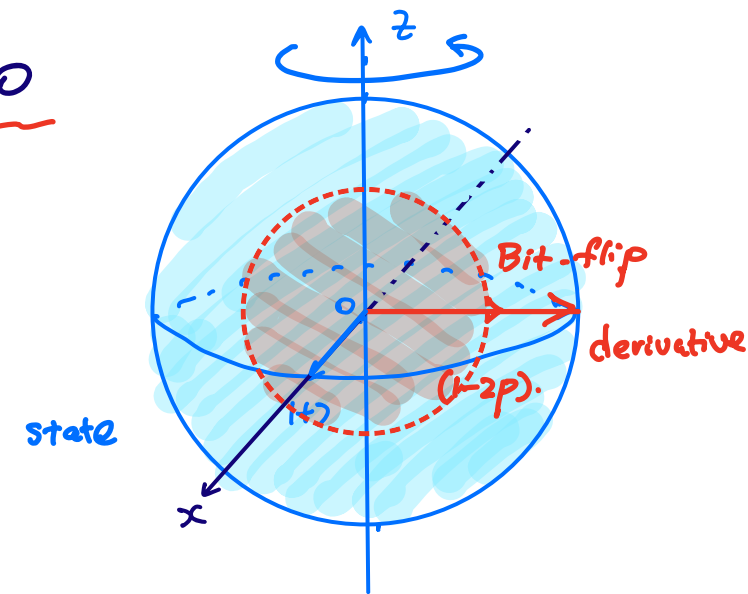
* : Bit-flip noise , calculate QFI at $\theta=0$

* : Input state : $|+\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$

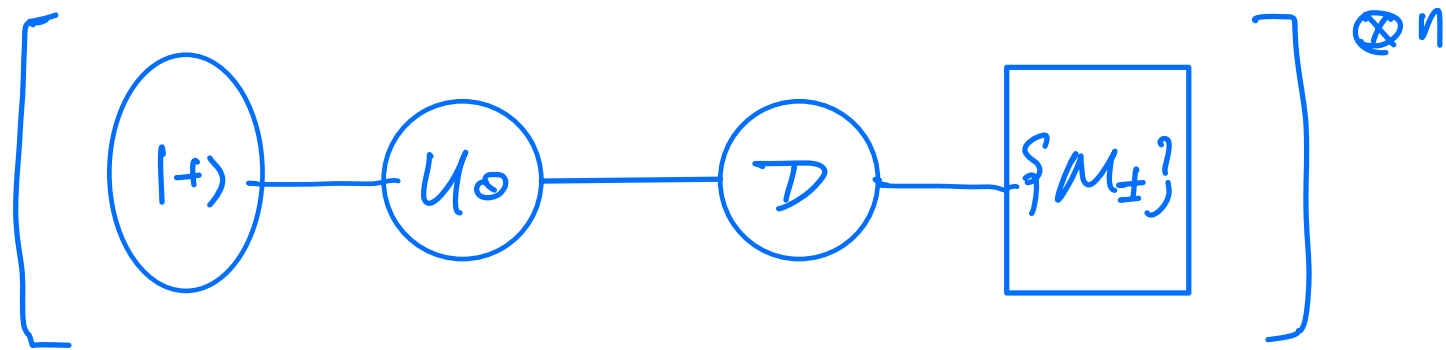
* : Output state : Bloch Representation

derivative w.r.t. θ $\rightarrow$ $(1, 0, 0) \xrightarrow{D} \dots \xrightarrow{D} (1, 0, 0)$
 $(1, 0, 0) \xrightarrow{U_0} (0, 1, 0) \xrightarrow{D} (0, 1-2p, 0) \rightarrow \dots \xrightarrow{D} (0, (1-2p)^n, 0)$

$$QFI = 4 \left[(1-2p) + \dots + (1-2p)^n \right]^2 \quad \text{----- approaching a constant}$$



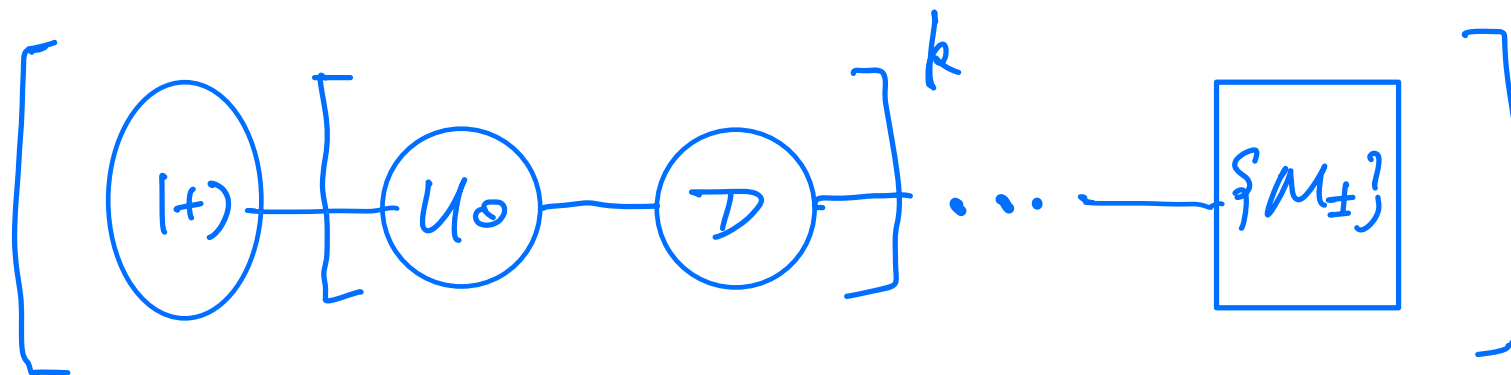
NOISY PHASE ESTIMATION — SEQUENTIAL



Bit-flip

measure n times

$$QFI = 4(1-2p)^2 n \sim SQL$$



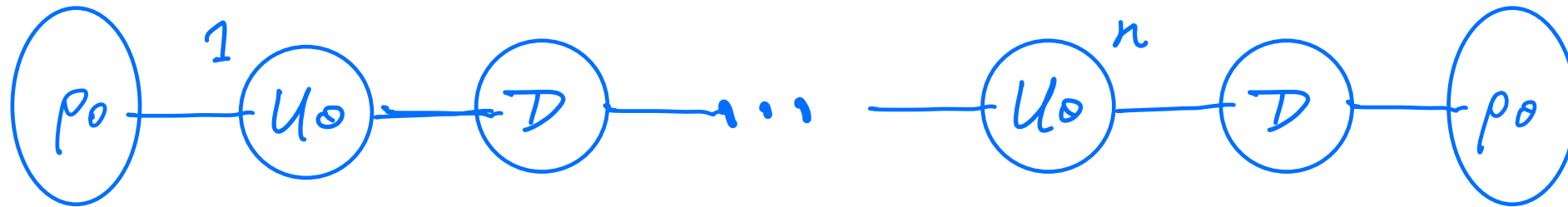
Bit-flip

measure n/k times

k is optimal at $\sim 0.6/p$

$$QFI \approx 0.8 n/p \sim SQL$$

NOISY PHASE ESTIMATION — SEQUENTIAL



* : Dephasing noise

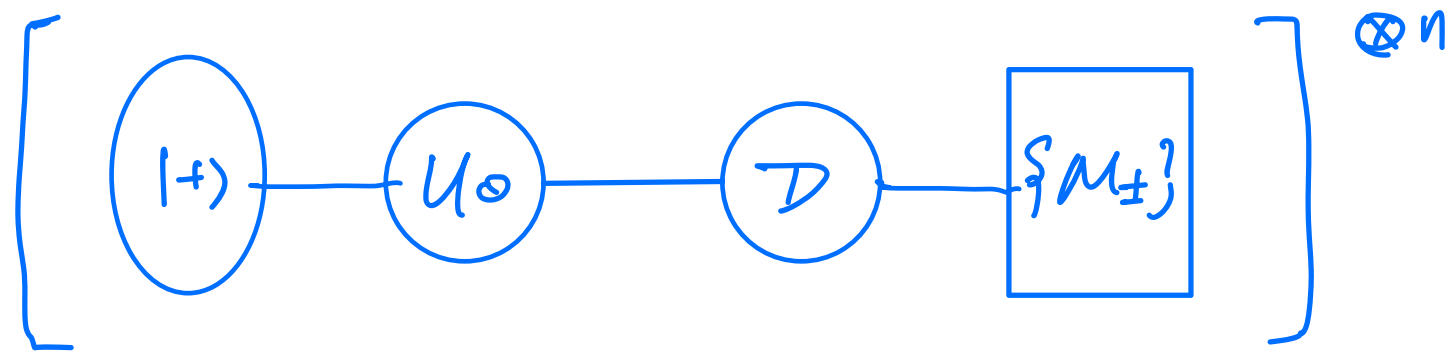
* : Input state : $|+\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$

* : Output state : $\rho_\theta = \frac{1}{2} \begin{bmatrix} 1 & (1-2p)^n e^{-2in\theta} \\ (1-2p)^n e^{2in\theta} & 1 \end{bmatrix}$

$$QFI = 4n^2 (1-2p)^{2n}$$

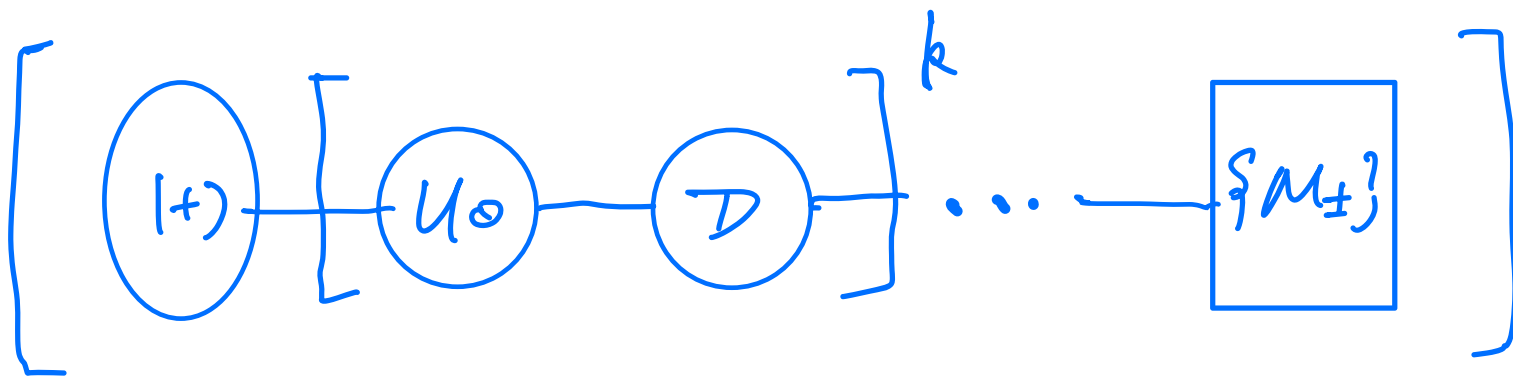
---- decays exponentially

NOISY PHASE ESTIMATION — SEQUENTIAL



Dephasing
measure n times

$$QFI = 4(1-2p)^2 n \quad \sim \text{SQL}$$



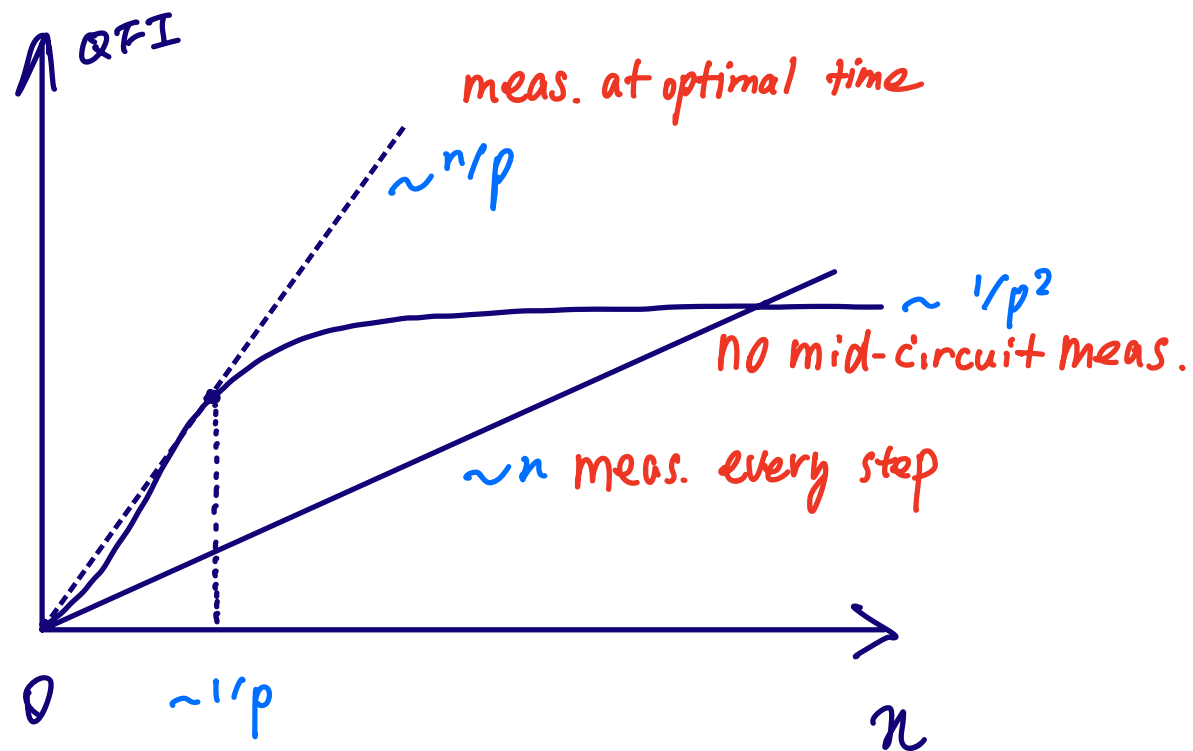
Dephasing
measure n/k times
 k is optimal at $\sim 0.25/p$

$$QFI \approx 0.37 \, n/p \quad \sim \text{SQL}$$

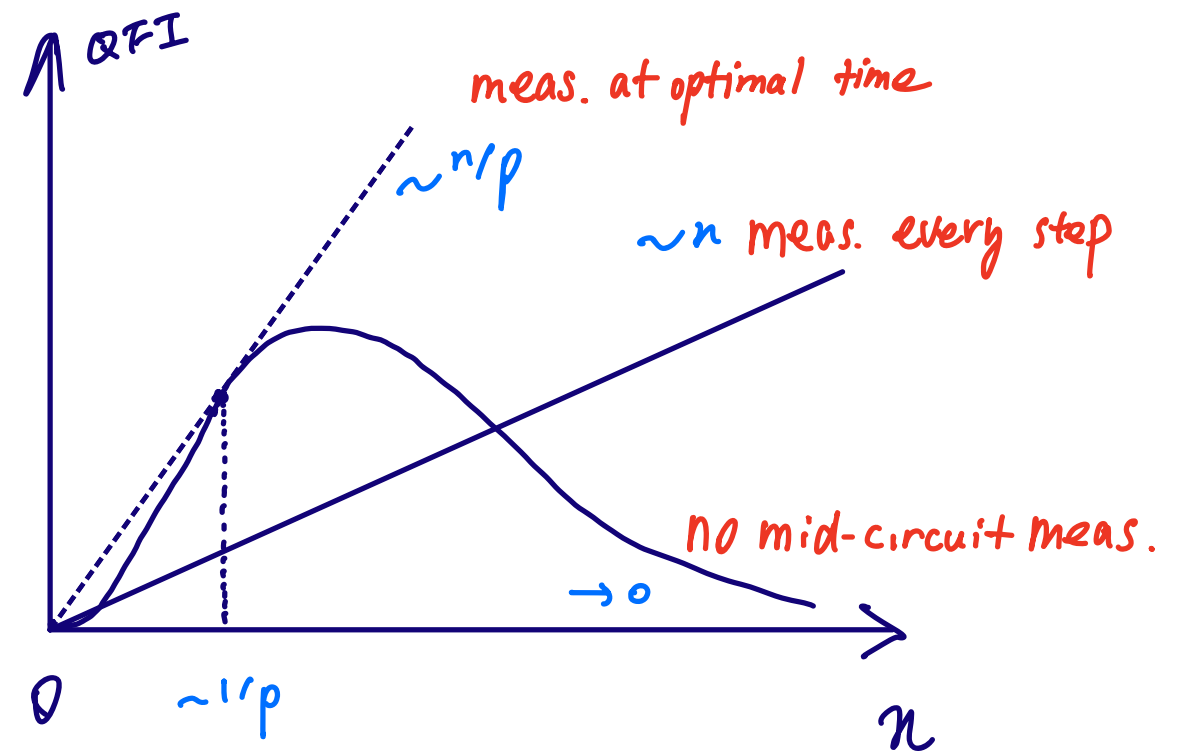
NOISY PHASE ESTIMATION — SEQUENTIAL

* input state: $|+\rangle$, with noise

BIT-FLIP — SQL



DEPHASING — SQL



SUMMARY OF PART 2

- * Definitions of HL, SQL, Parallel, Sequential
- * Phase estimation, Achieving the HL, GHZ state
- * Quantum Noise, Bit-flip, Dephasing noises
- * Noisy phase estimation, Achieving the HL in parallel strategy
- * In most cases, FI follows the SQL

Part 3

INTRODUCTION TO QUANTUM ERROR CORRECTION

* QUANTUM NOISE

* QUANTUM ERROR CORRECTION CONDITIONS

* Examples: $\left\{ \begin{array}{l} \text{Repetition Code} \\ \text{Shor Code} \\ \text{Surface code} \end{array} \right.$

QUANTUM NOISE

* $\mathcal{D}(\rho) = \sum_i K_i(\cdot) K_i^\dagger$, K_i : Kraus operators

s.t. $\sum_i K_i^\dagger K_i = \mathbb{I}$

CPTP map: $\text{Tr}(\mathcal{D}(\rho)) = \text{Tr}(\rho)$

$(\mathcal{D} \otimes \mathbb{I})(\rho) \geq 0$, if $\rho \geq 0$

* K_i : noise operators

i : indices of different types of noise

* Kraus representation is not unique $\begin{bmatrix} K_1' \\ \vdots \\ K_{n'}' \end{bmatrix} = u \begin{bmatrix} K_1 \\ \vdots \\ K_n \end{bmatrix}$

$u^\dagger u = \mathbb{I}_r$

QUANTUM NOISE

* Dephasing :

$$K_0 = \sqrt{1-p} \mathbb{I}$$
$$K_1 = \sqrt{p} Z$$

* Bit-flip :

$$K_0 = \sqrt{1-p} \mathbb{I}$$
$$K_1 = \sqrt{p} X$$

* Depolarizing :

$$K_0 = \sqrt{1-3p/4} \mathbb{I}$$
$$K_1 = \sqrt{p/4} X$$
$$K_2 = \sqrt{p/4} Y$$
$$K_3 = \sqrt{p/4} Z$$

QUANTUM ERROR CORRECTION CONDITION

* QUESTION: P is the projector onto code subspace,
"logical space"

Under what conditions, $\exists R$, decoding channel,

$R \circ D(p) = p$, for any $p = PpP$ p is in the code subspace
perfect recovery

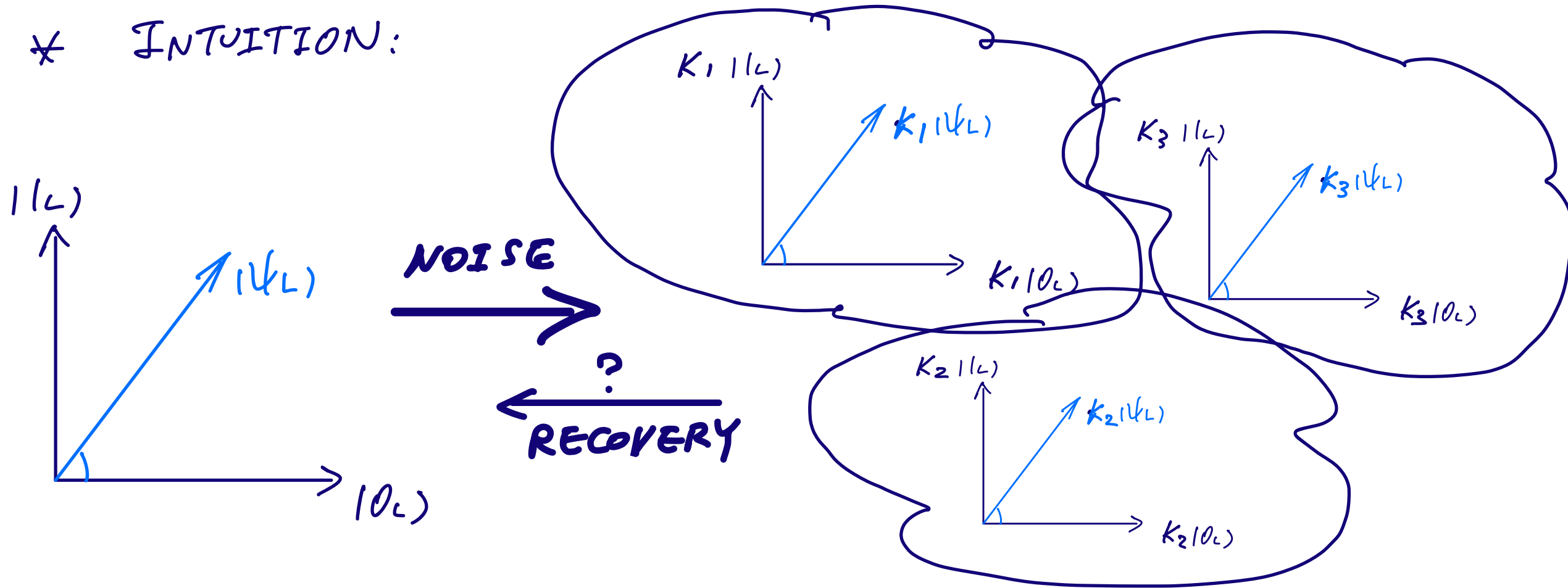
* ANSWER: The necessary & sufficient condition:

$$P K_i^\dagger K_j P = \alpha_{ij} P, \quad \forall i, j, \text{ some } \alpha_{ij} \in \mathbb{C}$$

Knill - Laflamme Condition

QUANTUM ERROR CORRECTION CONDITION

* INTUITION:



$$\langle 0_L | K^\dagger K | 1_L \rangle = 0 \text{ (orthogonal)}$$

$$\langle 0_L | K^\dagger K | 0_L \rangle = \langle 1_L | K^\dagger K | 1_L \rangle \text{ (same scale)}$$

$\Rightarrow$

$$P K^\dagger K P \propto P$$

QUANTUM ERROR CORRECTION CONDITION

* SUFFICIENCY

$$P K_i^\dagger K_j P = \alpha_{ij} P, \quad \forall i, j, \text{ some } \alpha_{ij} \in \mathbb{C}$$

$\alpha = u \lambda u^\dagger$, u is unitary, λ is diagonal

$\vec{F} = u^\dagger \vec{K}$ is another Kraus representation $P F_i^\dagger F_j P = \lambda_{ij} \delta_{ij} P$

$F_k P = \sqrt{\lambda_k} U_k P$ for some U_k , using polar decomposition.

$$P_k = U_k P U_k^\dagger, \quad R(\cdot) = \sum_k U_k^\dagger P_k(\cdot) P_k U_k$$

Detection P_k

Correction U_k

Satisfies $R \circ \mathcal{D}(\rho) = \rho$ for $P \rho P = \rho$.

QUANTUM ERROR CORRECTION CONDITION

* NECESSITY

$$\mathcal{R} \cdot \mathcal{P}(P\rho P) = P\rho P, \quad \forall \rho$$

$\{R_j\}$ and $\{P\}$ are Kraus operators of the same map,

$$\Rightarrow R_j P = c_j P \quad \text{for some } c_j$$

$$\Rightarrow P K_i^\dagger K_j P = P K_i^\dagger \left(\sum_k R_k^\dagger R_k \right) K_j P = \sum_k c_{ki}^* c_{kj} P = \alpha_{ij} P.$$

STABILIZER CODE

* n -qubit Pauli group: $\{I, X, Y, Z\}^{\otimes n} \times \{\pm 1, \pm i\}$

* Stabilizer group $\mathcal{S}$: subgroup of Pauli group satisfying
(1) $[S_i, S_j] = 0$, (2) $-I \notin \mathcal{S}$

* Stabilizer generators: $\mathcal{S} = \langle g_1, g_2, \dots, g_{n-k} \rangle$, $k \geq 1$

* Logical space: $\{|\psi\rangle : S|\psi\rangle = |\psi\rangle \forall S \in \mathcal{S}\}$ contains k logical qubits

*
$$P = \prod_{i=1}^{n-k} \frac{I + g_i}{2} = \frac{1}{2^{n-k}} \sum_{S \in \mathcal{S}} S$$

Sanity check:
$$SP = \frac{1}{2^{n-k}} \sum_{\underbrace{S' \in \mathcal{S}}} SS' = \frac{1}{2^{n-k}} \sum_{\underbrace{SS' \in \mathcal{S}}} SS' = P$$

 $\mathcal{S}$ is a group

STABILIZER CODE

* Consider Pauli noise $\mathcal{D}(\cdot) = \sum_i K_i(\cdot) K_i^\dagger$
where each K_i is proportional to a Pauli operator E_i

* QEC condition: $P E_i^\dagger E_j P \propto P$

✓ Situation 1: $E_i^\dagger E_j$ anti-commutes with some stabilizer g

$$P E_i^\dagger E_j g P = -P g E_i^\dagger E_j P, \quad gP = Pg = P$$
$$\Rightarrow P E_i^\dagger E_j P = -P E_i^\dagger E_j P \Rightarrow P E_i^\dagger E_j P = 0$$

✓ Situation 2: $E_i^\dagger E_j \propto$ a stabilizer $\Rightarrow P E_i^\dagger E_j P \propto P$

✗ Situation 3: $E_i^\dagger E_j$ commutes with all stabilizers, but is not proportional to one

If $P E_i^\dagger E_j P = \alpha P$, then $E_i^\dagger E_j |\psi_L\rangle = \alpha |\psi_L\rangle$ for all $|\psi_L\rangle$. $E_i^\dagger E_j / \alpha$ must be a stabilizer, (contradiction).

$E_i^\dagger E_j$ is called **LOGICAL** operators in situation 3.

STABILIZER CODE

* Example: Repetition Code

Stabilizers: $\langle z_1 z_2, z_2 z_3, \dots, z_{n-1} z_n \rangle = \langle z_i z_j \rangle$

Codewords: $|00 \dots 0\rangle, |1 \dots 1\rangle$

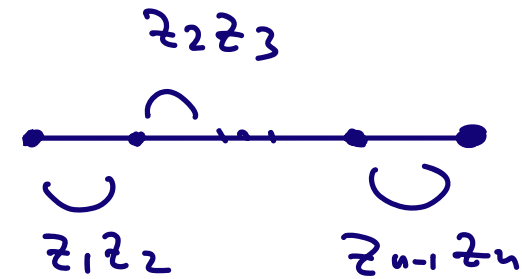
Errors: $x^{t_1} \otimes \dots \otimes x^{t_n}, |t| \leq \lfloor (n-1)/2 \rfloor$

why: $\begin{cases} E_1 = x^t, E_2 = x^s, |t \oplus s| \leq 2 \lfloor (n-1)/2 \rfloor < n \\ \checkmark \quad E_1^\dagger E_2 \text{ anticommutes with } z_i z_j, i \in t \cup s, j \notin t \cup s \end{cases}$

$\times \quad E_1 = z_1, E_2 = \mathbb{I}, E_1^\dagger E_2 = z_1, \text{ Commutes with } S \text{ but is not inside}$

Logical Operator: $z_L = z_i, x_L = x_1 x_2 \dots x_n$

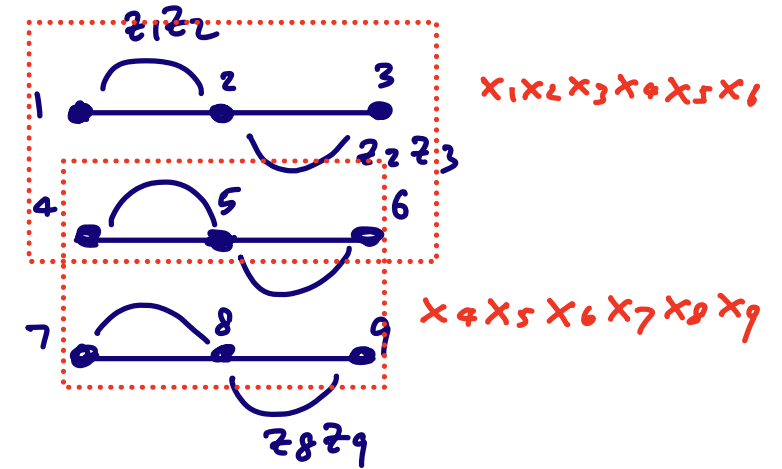
Decoding: Majority voting



STABILIZER CODE

* Example: Shor code

Stabilizers: $Z_1 Z_2, Z_2 Z_3, Z_4 Z_5, Z_5 Z_6, Z_7 Z_8, Z_8 Z_9$
 $X_1 X_2 X_3 X_4 X_5 X_6, X_4 X_5 X_6 X_7 X_8 X_9$



Codewords: $\left(\frac{|000\rangle \pm |111\rangle}{\sqrt{2}} \right)^{\otimes 3} = \begin{cases} |+_R +_R +_R\rangle \\ |-_R -_R -_R\rangle \end{cases}$

Errors: Single-qubit Errors

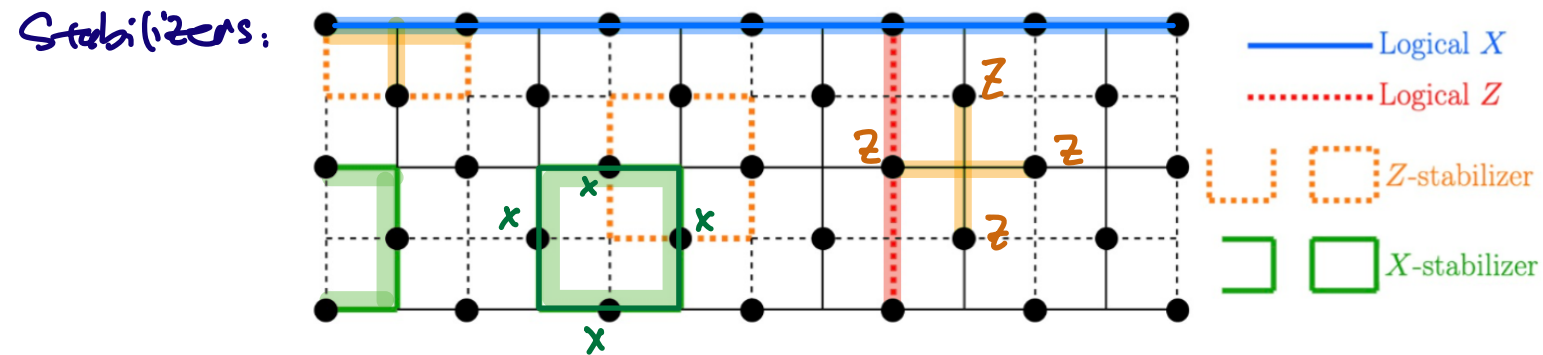
why: X noise, corrected by Z Stabs

Z noise, corrected by X Stabs.

Logical operation: $Z_L = X_1 X_2 X_3 X_4 X_5 X_6 X_7 X_8 X_9$
 $X_L = Z_1 Z_4 Z_7$

STABILIZER CODE

* Example : (Thin) Surface code



Errors: Single-qubit Errors

why : X noise , corrected by Z Stabs
 Z noise , corrected by X Stabs.

SUMMARY OF PART 3

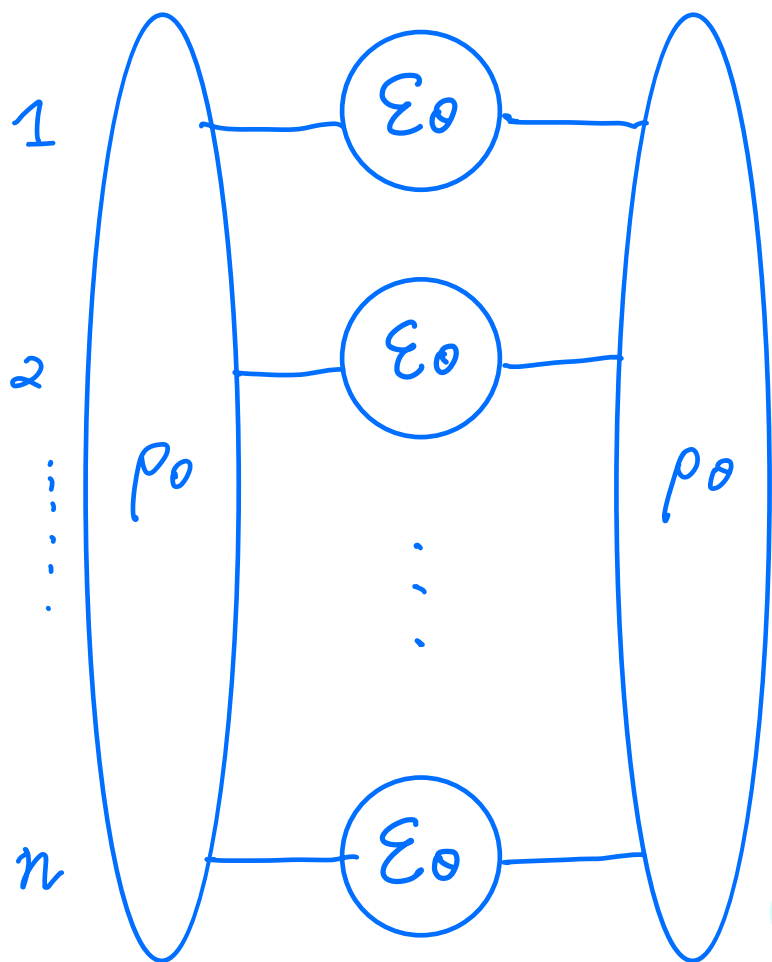
- * Definitions Quantum Noise (channel), Kraus Representation
- * QEC Condition ; Necessary and Sufficient
(KL)
- * Stabilizer Code ; QEC Condition
- * Example : Repetition code
Shor code (encoding one logical qubit)
Surface code

Part 4

QUANTUM ERROR CORRECTION FOR METROLOGY

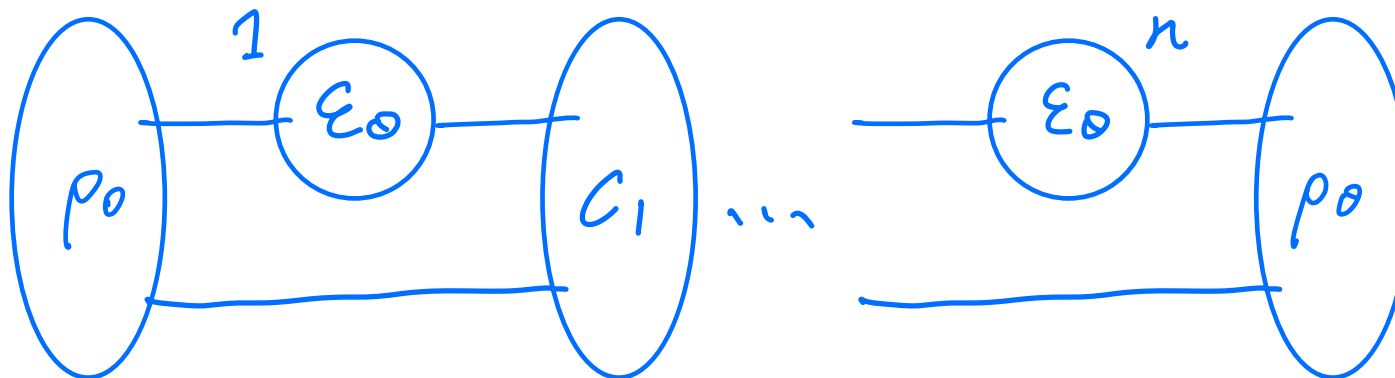
- * NOISY QUANTUM CHANNEL ESTIMATION
- * HAMILTONIAN NOT-IN - KRAUS-SPAN CONDITION
- * EXAMPLE: PHASE ESTIMATION UNDER BIT-FLIP NOISE
- * EXAMPLE: ZZZ ESTIMATION UNDER SINGLE-QUBIT NOISE

QUANTUM CHANNEL ESTIMATION



Parallel

max
all par(seq)
protocols



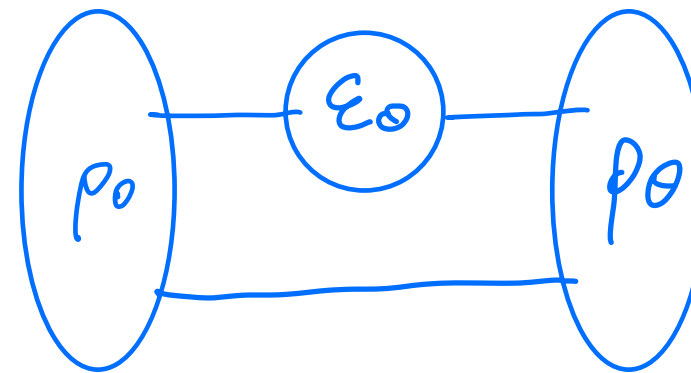
Sequential

$$F^{(par, seq)}(\rho_\theta) = \begin{cases} \sim n^2 & \text{HL} \\ \sim n & \text{SQL} \end{cases}$$

SINGLE-CHANNEL ESTIMATION

$$* \quad \varepsilon_{\theta}(\rho) = \sum_i K_{i,\theta}(\rho) K_{i,\theta}^{\dagger}$$

$$* \quad F(\Sigma_{\theta}) := \max_{\rho_0} F((\Sigma_{\theta} \otimes \mathbb{1})(\rho_0))$$



CHANNEL QFI

$$* \quad F(\Sigma_{\theta}) = 4 \min_{U: U^{\dagger} U = \mathbb{1}} \left\| \partial_{\theta} \vec{K}'^{\dagger} \cdot \partial_{\theta} \vec{K}' \right\| = \boxed{4 \min_h \|\alpha\|}$$

$$\vec{K}' = \begin{bmatrix} K'_{1,\theta} \\ \vdots \\ K'_{r,\theta} \end{bmatrix} = U_{\theta} \begin{bmatrix} K_{1,\theta} \\ \vdots \\ K_{r,\theta} \end{bmatrix}, \quad h = i U_{\theta}^{\dagger} (\partial_{\theta} U_{\theta}), \quad \alpha = (\partial_{\theta} \vec{K} - i h \vec{K})^{\dagger} (\partial_{\theta} \vec{K} - i h \vec{K})$$

minimize over all possible Kraus representation $\Leftrightarrow$ minimize over all Hermitian h

MULTI-CHANNEL ESTIMATION

$$* \quad F_n^{(par)} \leq 4 \min_h \left(\overset{SQL}{\|\alpha\| n} + \overset{HL}{\|\beta\|^2 n(n-1)} \right)$$

$$* \quad F_n^{(seq)} \leq 4 \min_h \left(\|\alpha\| n + \|\beta\| (\|\beta\| + 2\sqrt{\|\alpha\|}) n(n-1) \right)$$

$$\begin{cases} \alpha = (\partial_\theta \vec{k}_\theta - i h \vec{k}_\theta)^\dagger (\partial_\theta \vec{k}_\theta - i h \vec{k}_\theta) \\ \beta = i \vec{k}_\theta^\dagger (\partial_\theta \vec{k}_\theta - i h \vec{k}_\theta) = i \vec{k}_\theta^\dagger (\partial_\theta \vec{k}_\theta) + \vec{k}_\theta^\dagger h \vec{k}_\theta =: \underbrace{H_\theta + \vec{k}_\theta^\dagger h \vec{k}_\theta}_{\text{Hamiltonian}} \end{cases}$$

$$* \quad \boxed{\exists h, \quad \beta = 0} \Rightarrow F_n^{(par)} \leq F_n^{(seq)} \leq O(n)$$



$$\boxed{H_\theta \in \text{span} \{ K_{i,\theta}^\dagger K_{j,\theta}, \forall i,j \}}$$

"Hamiltonian-in-Kraus-span"

THEOREM STATEMENT

* $\exists$ parallel (or sequential) strategy, s.t.

$$F^{(\text{par})} \text{ (or } F^{(\text{seq})}) = \Theta(n^2)$$

IF AND ONLY IF

the HNKS condition is satisfied.

$$H_0 \notin \text{span} \{ K_{i,0}^\dagger K_{j,0}, \forall i,j \}$$

"Hamiltonian-not-in-Kraus-span"
HNKS Condition

* If HNKS violated, $F^{(\text{par}, \text{seq})} = \mathcal{O}(n)$, (SQL)

SUFFICIENCY: QEC PROTOCOL.

* QEC condition: $P K_{i,0}^\dagger K_{j,0} P \propto P \quad \forall i, j$

* Goal: Find a QEC code P , s.t. $\begin{cases} P K_{i,0}^\dagger K_{j,0} P \propto P \\ P H_0 P \not\propto P \end{cases}$

$\exists$ Recovery R , s.t.

$$\rho_{\theta+d\theta} = R \circ \Sigma_{\theta+d\theta}(\rho) = \rho_\theta - i [\underbrace{P H_0 P}_{H_L}, \rho_\theta] d\theta + O(d\theta^2)$$

Assume $P K_i^\dagger K_j P = \chi_{ij} \delta_{ij} P$ (which can be achieved changing the Kraus rep)

Then let $K_i P = \sqrt{\chi_i} U_i P$, $P_i = U_i P U_i^\dagger$, $R = \sum_i U_i^\dagger P_i(\cdot) P_i U_i = \sum_i \underbrace{P U_i^\dagger(\cdot) U_i P}_{\text{blue wavy}}$

$$\Sigma_{\theta+d\theta}(\rho) - \Sigma_\theta(\rho) = \left[\sum_i (\partial_\theta K_{i,0}) P K_{i,0}^\dagger + \sum_i K_{i,0} P (\partial_\theta K_{i,0}^\dagger) \right] d\theta + O((d\theta)^2)$$

SUFFICIENCY: QEC PROTOCOL. (CONT'D)

$R \circ \Sigma_\theta(p) = p$ due to the QEC condition

$$p = p p p = p p$$

$$R(\Sigma_\theta + d\theta(p) - \Sigma_\theta(p)) = \left[\sum_i \underbrace{p}_{\text{blue}} \underbrace{v_i^\dagger}_{\text{blue}} (\partial_\theta k_{i,\theta}) \underbrace{p}_{\text{red}} \underbrace{p}_{\text{red}} \underbrace{k_i^\dagger}_{\text{blue}} \underbrace{v_i}_{\text{blue}} p + \text{h.c.} \right] d\theta + O(d\theta^3)$$

$$= \left[\sum_i \underbrace{p}_{\text{blue}} \underbrace{v_i^\dagger}_{\text{blue}} (\partial_\theta k_{i,\theta}) \underbrace{p}_{\text{red}} \underbrace{p}_{\text{red}} \underbrace{v_i^\dagger \bar{\lambda}_i v_i}_{\text{blue}} p + \text{h.c.} \right] d\theta + O(d\theta^3)$$

$$= \left[\sum_i \underbrace{p}_{\text{blue}} \underbrace{v_i^\dagger \bar{\lambda}_i}_{\text{blue}} (\partial_\theta k_{i,\theta}) \underbrace{p}_{\text{red}} p + \text{h.c.} \right] d\theta + O(d\theta^3)$$

$$= \left[\sum_i \underbrace{p}_{\text{blue}} \underbrace{k_i^\dagger}_{\text{red}} (\partial_\theta k_{i,\theta}) \underbrace{p}_{\text{red}} p + \text{h.c.} \right] d\theta + O(d\theta^3)$$

$-iH_\theta$

$$= p [-i[H_\theta, p]] p d\theta + O(d\theta^3)$$

$$= -i \left[\underbrace{p H_\theta p}_{\text{red}}, p \right] d\theta + O(d\theta^2)$$

$H_L \neq I_L$

SUFFICIENCY: QEC PROTOCOL. (CONT'D)

* We need to find P s.t.

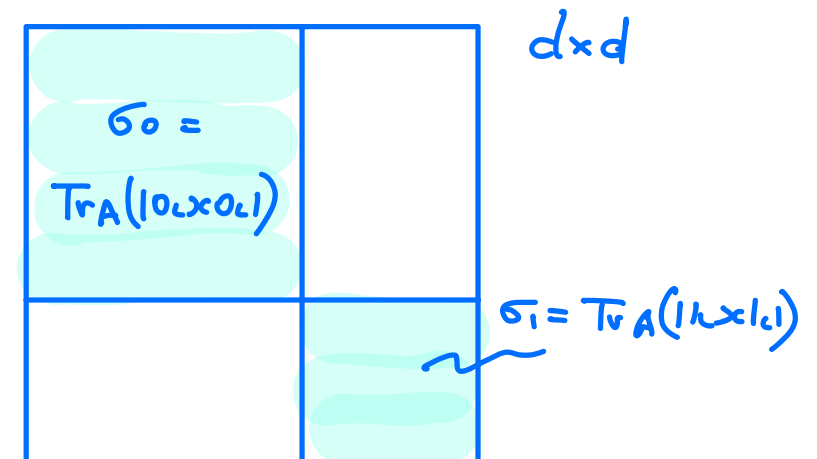
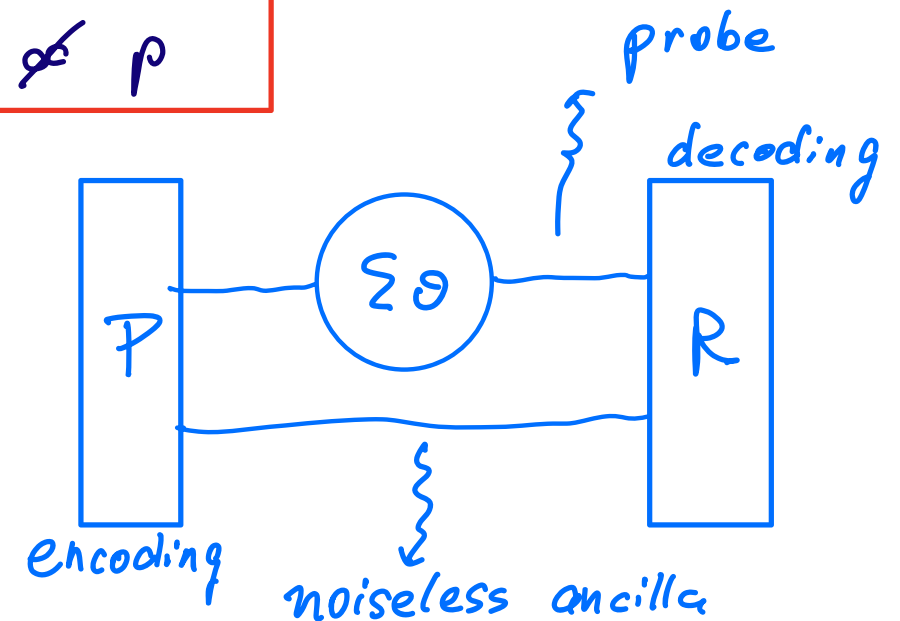
$$\begin{cases} P K_{i,0}^\dagger K_{j,0} P \propto P \\ P H_0 P \neq P \end{cases}$$

* Noiseless ancilla helps:

$$\begin{cases} \langle 0_L | K_{i,0}^\dagger K_{j,0} \otimes \mathbb{1}_A | 1_L \rangle = 0 & \forall i,j \\ \langle 0_L | K_{i,0}^\dagger K_{j,0} \otimes \mathbb{1}_A | 0_L \rangle = \langle 1_L | K_{i,0}^\dagger K_{j,0} \otimes \mathbb{1}_A | 1_L \rangle \\ \langle 0_L | H_0 \otimes \mathbb{1}_A | 0_L \rangle \neq \langle 1_L | H_0 \otimes \mathbb{1}_A | 1_L \rangle \end{cases}$$

$$\begin{aligned} \Leftarrow \quad & \text{Tr}((\sigma_0 - \sigma_1) K_{i,0}^\dagger K_{j,0}) = 0 \quad \forall i,j \\ & \text{Tr}((\sigma_0 - \sigma_1) H_0) \neq 0 \end{aligned}$$

$$\Leftarrow \quad H_0 \notin \text{span} \{ K_{i,0}^\dagger K_{j,0}, \forall i,j \}$$



THEOREM STATEMENT — REVISITED

* $\exists$ parallel (or sequential) strategy, s.t.

$$F^{(\text{par})} \text{ (or } F^{(\text{seq})}) = \Theta(n^2)$$

IF AND ONLY IF $H_0 \notin \text{span} \{ K_{i,0}^\dagger K_{j,0}, \forall i,j \}$ (HNKS)

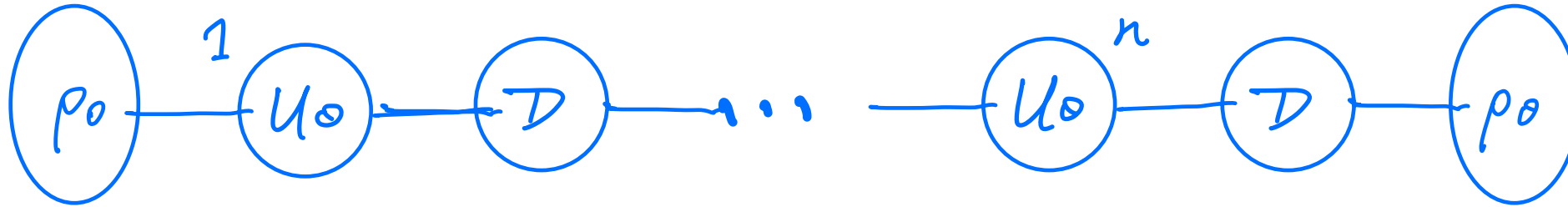
* NECESSITY: $F \leq \|\alpha\| O(n) + \|\beta\| O(n^2)$

$$\nexists h, \beta=0 \Leftrightarrow \text{HNKS}$$

* SUFFICIENCY: HNKS + Noiseless ancilla $\Rightarrow$

$$\exists P, \text{ s.t. } \begin{cases} P K_{i,0}^\dagger K_{j,0} P \propto P \\ P H_0 P \neq P \end{cases} \Rightarrow \text{Effective unitary evolution}$$

NOISY PHASE ESTIMATION UNDER BIT-FLIP NOISE



* : Bit-flip noise

* : Input state : $|+\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$

QFI = $O(1)$, if no control

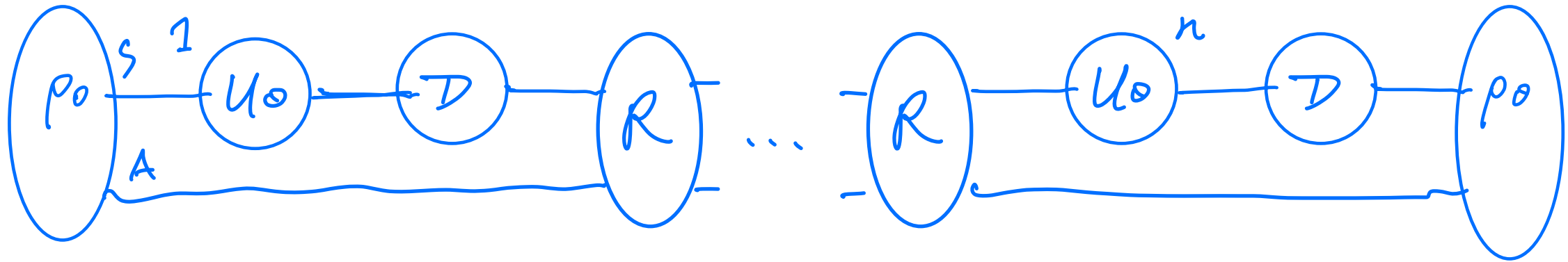
* HNKS: $\Sigma_\theta = \mathcal{D} \circ U_\theta$, $\begin{cases} K_1 = \sqrt{1-p} e^{-iZ\theta} \\ K_2 = \sqrt{p} X e^{-iZ\theta} \end{cases}$

Hamiltonian: $H = i K_1^\dagger \partial_\theta K_1 + i K_2^\dagger \partial_\theta K_2 = K_1^\dagger K_1 \cdot Z + K_2^\dagger K_2 \cdot Z = \underline{Z}$

Kraus span: $\text{span} \{ \mathbb{I}, X \}$ ✓ HNKS satisfied, HL achievable

For dephasing noise, however, $Z \in \text{span} \{ \mathbb{I}, Z \}$ ✗ HNKS violated, HL not achievable

NOISY PHASE ESTIMATION UNDER BIT-FLIP NOISE



* QEC is possible, using noiseless ancilla

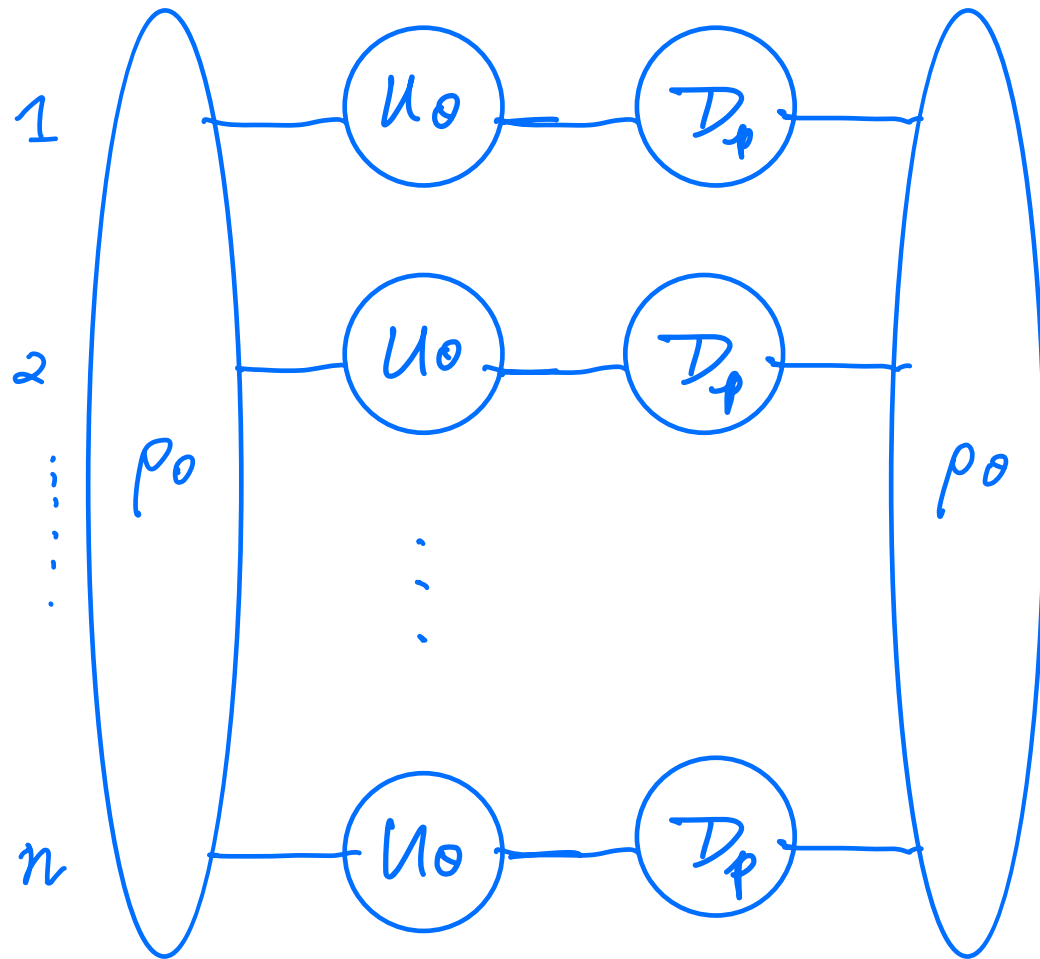
* Repetition code is sufficient:

$$|0_L\rangle = |0_S\rangle|0_A\rangle, \quad |1_L\rangle = |1_S\rangle|1_A\rangle$$

* Detection: $\begin{cases} P_1 = |00\rangle\langle 00| + |11\rangle\langle 11| & \dots \text{No. error} \\ P_2 = |01\rangle\langle 01| + |10\rangle\langle 10| & \dots \text{bit-flip error on } S \end{cases}$

* Correction: $\begin{cases} U_1 = I \otimes I \\ U_2 = X \otimes I \end{cases}$ * $R[(D \circ U_0) \otimes I_A] \sim U_0 \otimes I_A$,
in the code subspace.

NOISY PHASE ESTIMATION UNDER BIT-FLIP NOISE



* n -qubit GHZ state + Bit-flip noise
 $\text{QFI} = 4n^2$, as in the noiseless case

* QEC interpretation [with caveat]

* Repetition Code:

$$|0_c\rangle = |00 \dots 0\rangle \quad |1_c\rangle = |11 \dots 1\rangle$$

* QEC condition:

Errors: $x^{t_1} \otimes \dots \otimes x^{t_n}$, $|t| \leq \lfloor (n-1)/2 \rfloor$

We want $\langle 0 \dots 0 | K_i^\dagger K_j | 1 \dots 1 \rangle = 0$

$\langle 0 \dots 0 | x^{s_1} \otimes \dots \otimes x^{s_n} | 1 \dots 1 \rangle = 0$ if $\text{weight}(s) \leq n-1$,

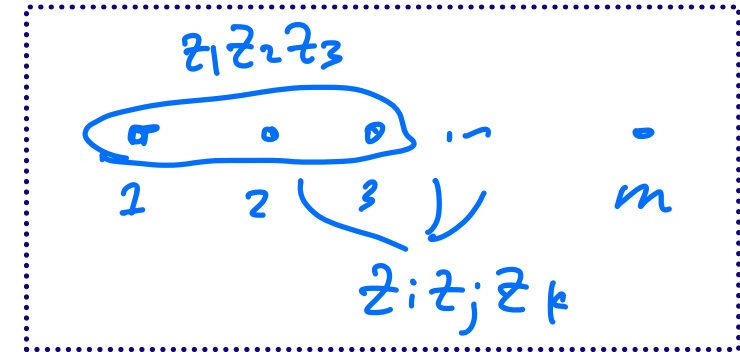
* We can tolerate $\lfloor (n-1)/2 \rfloor$ number of bit-flip noises

* The probability of getting $> \lfloor (n-1)/2 \rfloor$ number of noises is exponentially small in n (when $p < 1/2$)

ZZZ INTERACTION UNDER SINGLE-QUBIT NOISE

$$* U_\theta = \exp(-iH\theta),$$

$$H = \sum_{ijk} Z_i \otimes Z_j \otimes Z_k, \text{ } ijk \text{ distinct}$$



code block

$$\#(ijk): \binom{m}{3} = \frac{m(m-1)(m-2)}{6}$$

$$* \mathcal{D}(\cdot) = \sum_i p_{i,x} x_i(\cdot) x_i + \sum_i p_{i,y} y_i(\cdot) y_i + \sum_i p_{i,z} z_i(\cdot) z_i$$

not i.i.d. noise

ground state
↙

highest excited state
↘

$$* \text{Noiseless case: Use GHZ: } \frac{|0 \dots 0\rangle + |1 \dots 1\rangle}{\sqrt{2}}$$

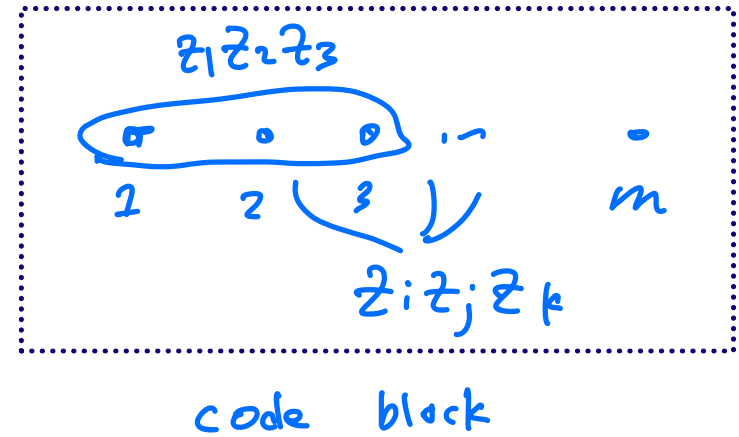
$$Z_i Z_j Z_k |0 \dots 0\rangle = |0 \dots 0\rangle; \quad Z_i Z_j Z_k |1 \dots 1\rangle = -|1 \dots 1\rangle.$$

$$\underline{QFI = 4 \cdot \binom{m}{3}^2 \cdot n^2}$$

ZZZ INTERACTION UNDER SINGLE-QUBIT NOISE

* $U_\theta = \exp(-iH\theta)$,
 $H = \sum_{i,j,k} Z_i \otimes Z_j \otimes Z_k$, i,j,k distinct

#(ijk): $\binom{m}{3} = \frac{m(m-1)(m-2)}{6}$



* Noisy case: $\{|0\dots 0\rangle, |1\dots 1\rangle\}$ is not a good QEC code,
 e.g. $\langle 0\dots 0 | Z | 0\dots 0 \rangle \neq \langle 1\dots 1 | Z | 1\dots 1 \rangle$

* Use (generalized) Shor code

Codewords: $\left(\frac{|000\rangle \pm |111\rangle}{\sqrt{2}} \right)^{\otimes 3} \longrightarrow \left(\frac{|0\dots 0\rangle \pm |1\dots 1\rangle}{\sqrt{2}} \right)^{\otimes 3} =: \begin{cases} |+_R+_R+_R\rangle \\ |-_R-_R-_R\rangle \end{cases}$

Shor code (generalized) Shor code :

ZZZ INTERACTION UNDER SINGLE-QUBIT NOISE

* $\left(\frac{|0\dots 0\rangle \pm |1\dots 1\rangle}{\sqrt{2}} \right)^{\otimes 3} \xrightarrow{\begin{matrix} X \text{ noise} \\ Z \text{ noise} \end{matrix}} \left(\frac{|010\dots 0\rangle \pm |101\dots 1\rangle}{\sqrt{2}} \right) \left(\frac{|10\dots 0\rangle \pm |11\dots 1\rangle}{\sqrt{2}} \right)^{\otimes 2}$

Can correct single-qubit noise

$\xrightarrow{\text{noise}} |-\rangle_R |+\rangle_R |+\rangle_R$

* QFI:

1	$m/3$	$m/3+1$	$2m/3$	$2m/3+1$	m	Logical / Noise
$(\dots 0)$	$(\dots 0)$	$(\dots 0)$	$(\dots 0)$	$(\dots 0)$	$(\dots 0)$	
Z_i		Z_j		Z_k		X_L
$Z_i Z_j$		Z_k				= single Z noise
$Z_i Z_j Z_k$						= single Z noise

* QFI: $4n^2 \cdot \left(\frac{n}{3}\right)^6 < 4n^2 \left(\frac{n}{3}\right)^3$

$= O(n^2 m^6)$

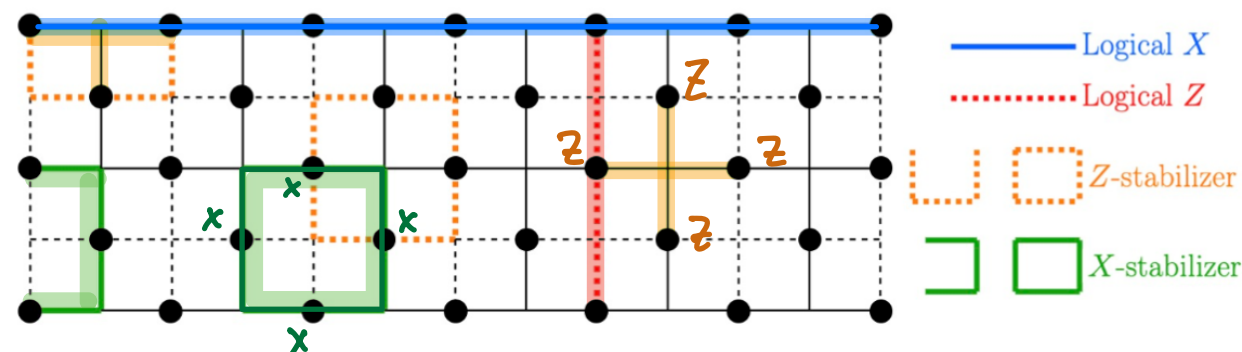
noiseless QFI

* Recover noiseless QFI up to constant factor

ZZZ INTERACTION UNDER SINGLE-QUBIT NOISE

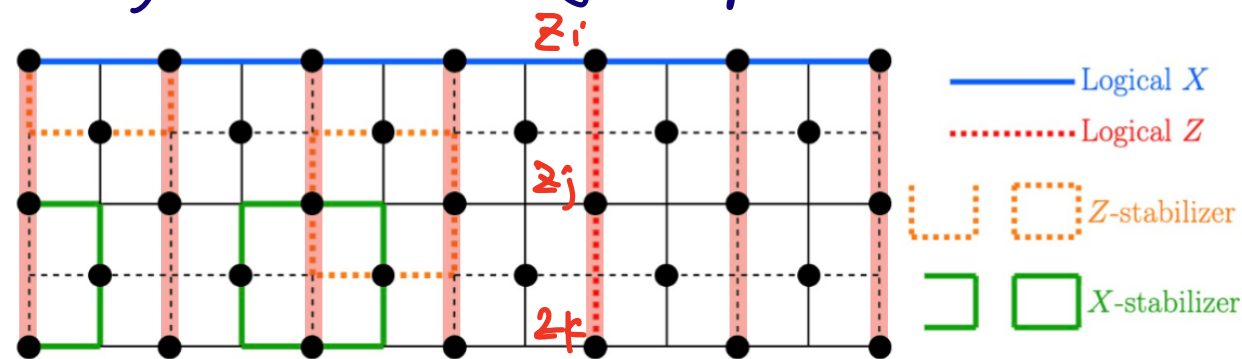
* Use thin surface code

Stabilizers:



Any $E_i^\dagger E_j$, where $E_{i,j}$ is single-qubit. Anti-commutes with some stabilizer.

* Number of $Z_i Z_j Z_k$ that is logical operation: $\sim m$



* $QFI \sim O(n^2 m^2) < O(n^2 m^6)$, [sub optimal factor]

SUMMARY OF PART 4

- * HKS condition $\Leftrightarrow T = \Theta(n^2)$
- * Necessity: Channel CFI upper bounds
- * Sufficiency: QEC protocol using noiseless ancilla
- * phase estimation under bitflip noise : repetition code
 - Full recover noiseless behaviour
- * ZZZ interaction within m -qubit code blocks
 - Repetition code fails to achieve HL
 - Generalized Shor code recovers optimal m -scaling
 - Thin surface code recovers suboptimal m -scaling

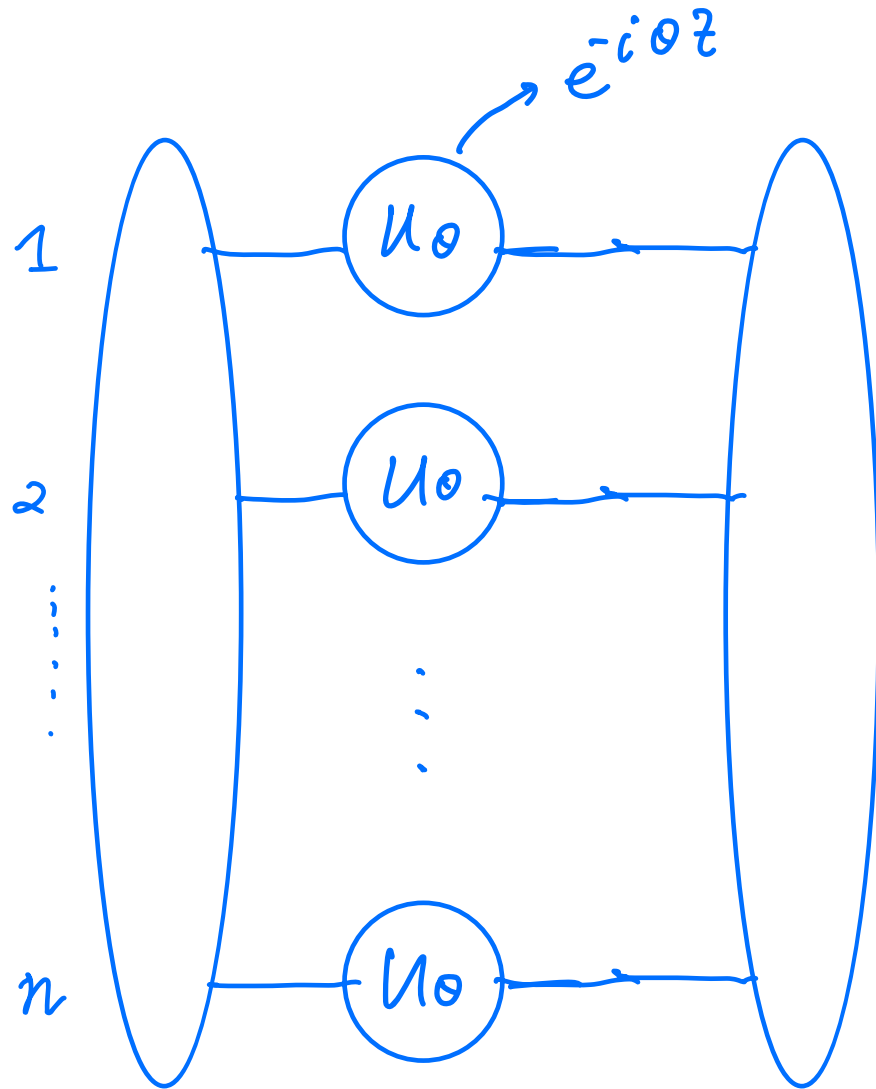
PART 5

FAULT TOLERANCE IN QUANTUM METROLOGY

—— PHASE ESTIMATION UNDER BIT-FLIP NOISE

- * SETTING
- * MEASUREMENT
- * STATE PREPARATION
- * Plot of Fisher information

SETTING



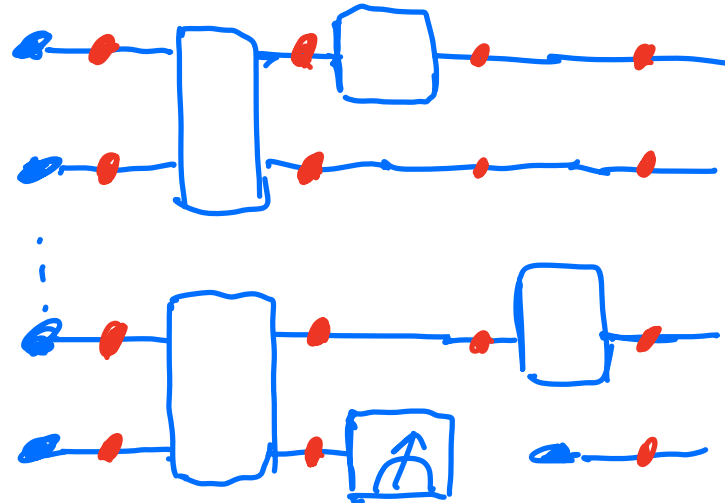
* CIRCUIT-LEVEL NOISE

- Initialization : $|+\rangle, |0\rangle$

- Measurement : $\{|0\rangle\langle 0|, |1\rangle\langle 1|\}$

- Gate and idle error: Bit-flip noise, X

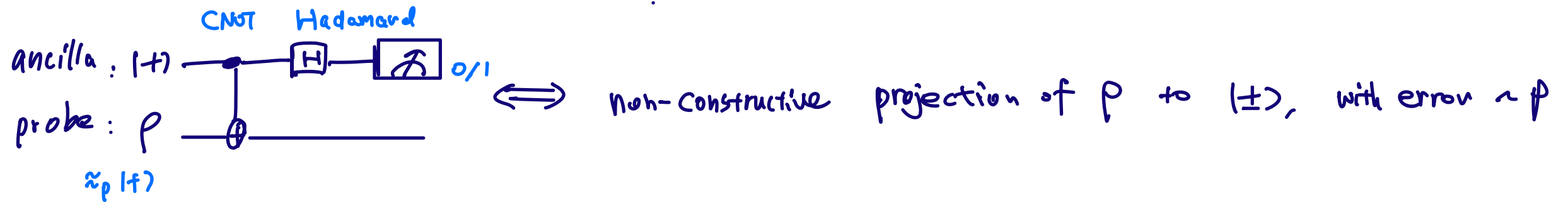
* NOISE PROBABILITY: p



STATE PREPARATION — SINGLE QUBIT INITIALIZATION

* Goal: Preparing $|+\rangle^{\otimes n}$

* Single-qubit Initialization error:



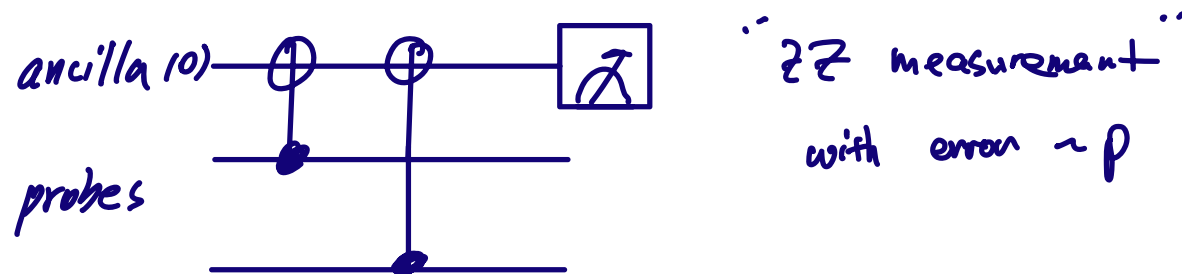
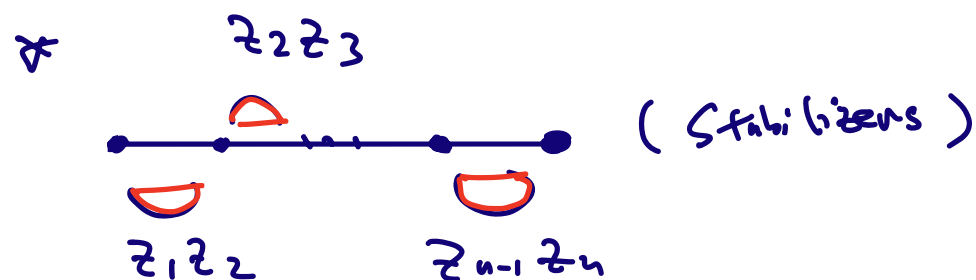
$$\begin{aligned} \text{CNOT } |0\rangle|+\rangle &\rightarrow |0\rangle|+\rangle, & \text{CNOT } |+\rangle|+\rangle &\rightarrow |+\rangle|+\rangle, & H|0\rangle &\rightarrow (|0\rangle + |1\rangle)/\sqrt{2} \\ \text{CNOT } |1\rangle|+\rangle &\rightarrow |1\rangle|+\rangle, & \text{CNOT } |+\rangle|-\rangle &\rightarrow |-\rangle|+\rangle, & H|1\rangle &\rightarrow (|0\rangle - |1\rangle)/\sqrt{2} \end{aligned}$$

Outcome 0 if ρ is $|+\rangle$, 1 if ρ is $|-\rangle$.

* Repeat measurement $\sim O(\log n)$ times, error prob $\xrightarrow{n} 0$

STATE PREPARATION — LOGICAL-STATE PREPARATION

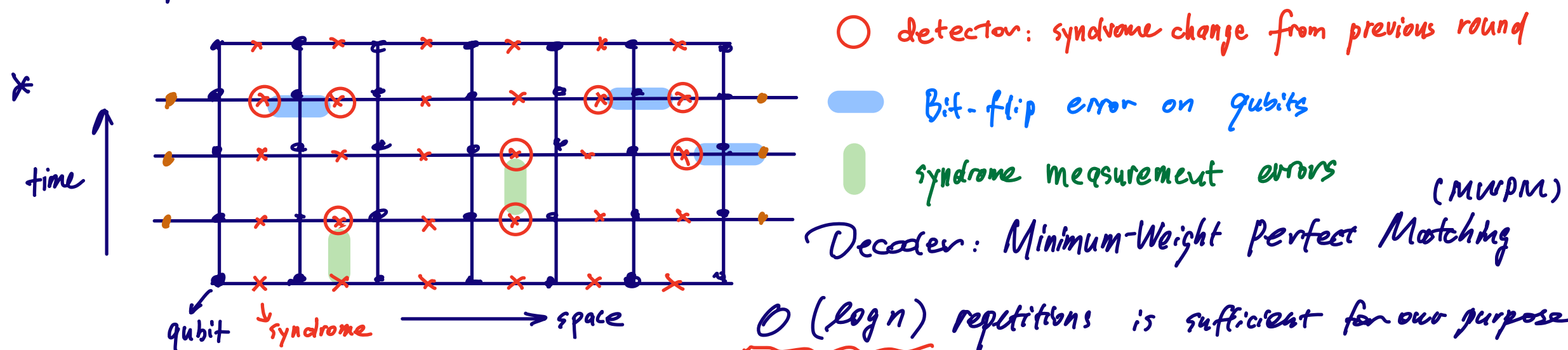
* Goal: Preparing $(|0\rangle^{\otimes n} + |1\rangle^{\otimes n}) / \sqrt{2}$ from $|+\rangle^{\otimes n}$



* Start with $|+\rangle^{\otimes n}$, Measure $z_1 z_2, z_2 z_3, \dots, z_{n-1} z_n$, Perform Correction

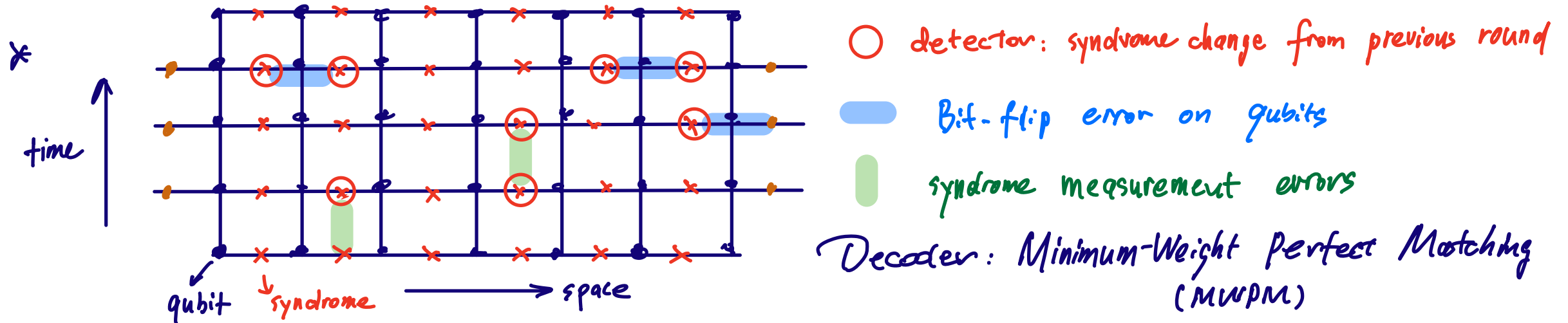
$\Rightarrow$ Output state must be logical state, stabilized by $X_L = x_1 x_2 \dots x_n$

$\Rightarrow$ Output state must be $GHZ = (|0\rangle + |1\rangle) / \sqrt{2}$



STATE PREPARATION — LOGICAL-STATE PREPARATION (CONT'D)

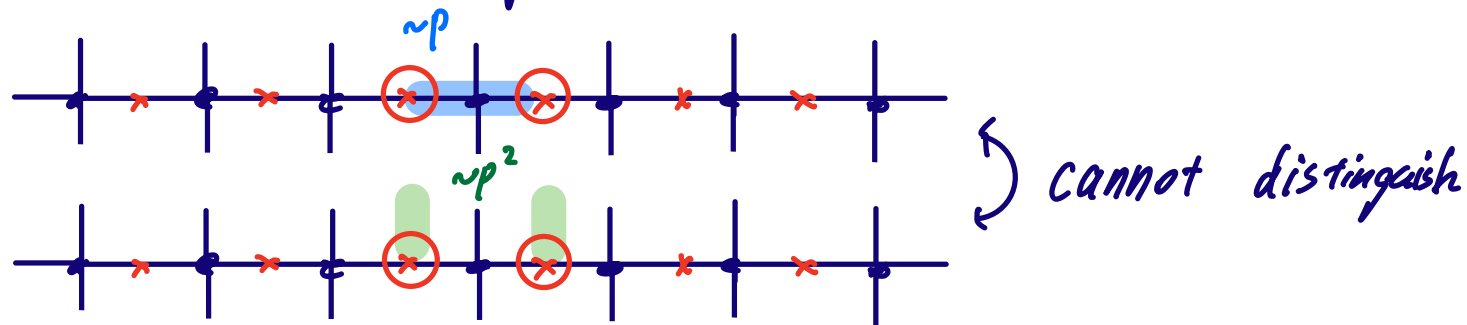
* Goal: Preparing $(|0\rangle^{\otimes n} + |1\rangle^{\otimes n}) / \sqrt{2}$ from $|+\rangle^{\otimes n}$



* $(|\vec{0} \oplus \vec{b}\rangle + |\vec{1} \oplus \vec{b}\rangle) / \sqrt{2}$

Magnetization: $n - 2 \text{wt}(b)$, b : # of bit flip errors

* errors happening in the end cannot fully corrected, $\text{wt}(b) \sim n \cdot p^2$



STATE PREPARATION ——— S GATE

$$* \frac{|0^{\otimes n}\rangle + |1^{\otimes n}\rangle}{\sqrt{2}} \longrightarrow \frac{|0^{\otimes n}\rangle + i|1^{\otimes n}\rangle}{\sqrt{2}}$$

* Perform S gate $\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$ randomly on a single qubit

* Reason: (Assume no error) When $\theta \approx 0$,

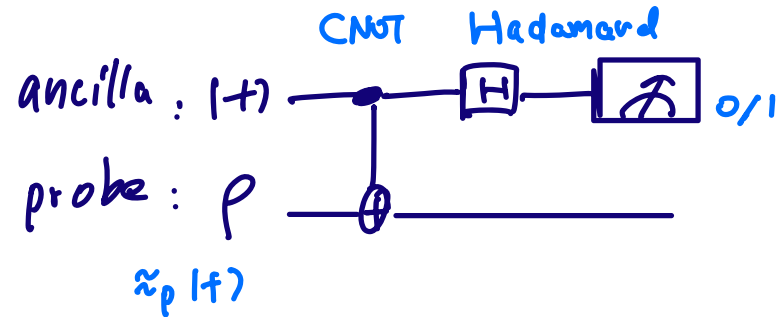
Before S gate : $p(\text{even}) = \cos^2(2n\theta) \approx 1$, $p(\text{odd}) = \sin^2(2n\theta) \approx 0$

After S gate : $p(\text{even}) = \frac{1 - \sin(2n\theta)}{2} \approx 1/2$, $p(\text{odd}) = \frac{1 + \sin(2n\theta)}{2} \approx 1/2$

We want a balanced probability distribution, which is more robust against measurement noise

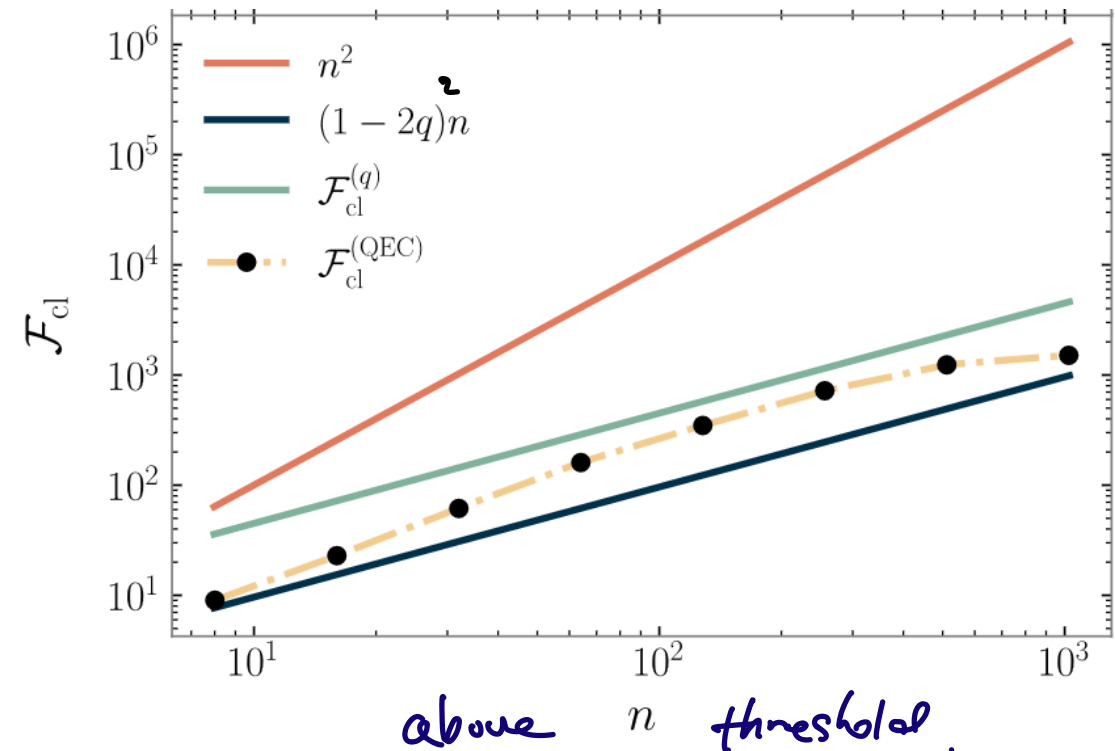
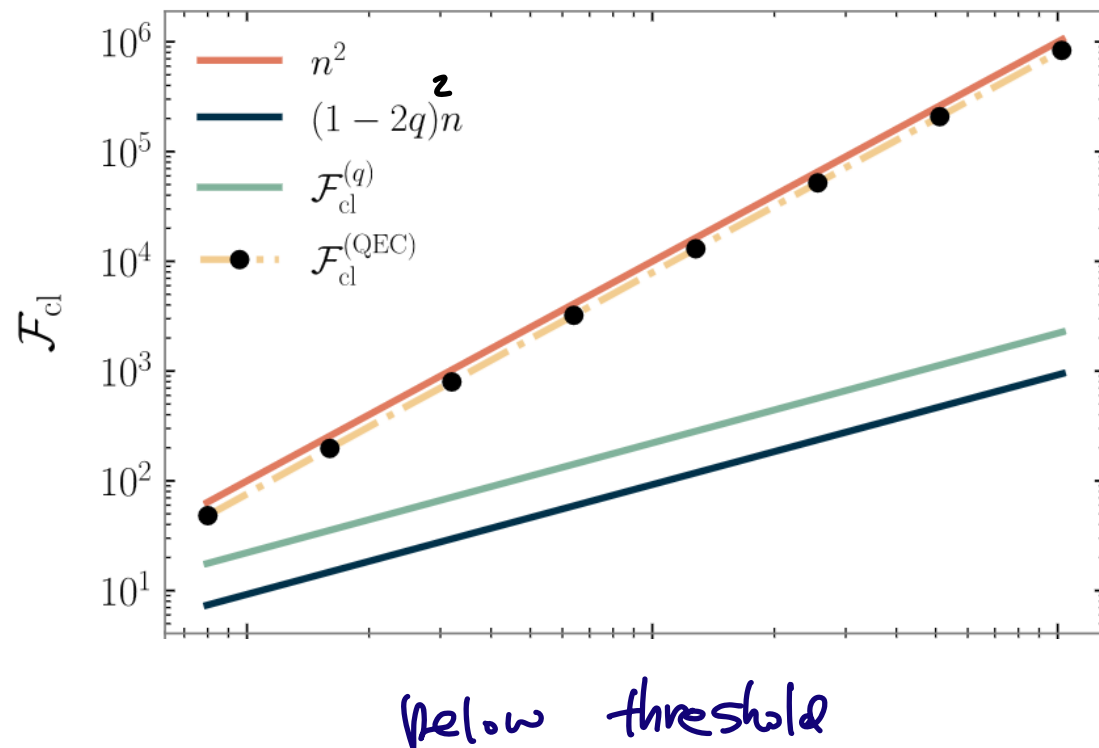
MEASUREMENT NOISE + PLOT OF FI

* Non-destructive X measurement, repeat $O(\log n)$ times on all qubits



$\Leftrightarrow$ non-destructive projection of ρ to $|\pm\rangle$, with error $\sim p$

* FI Plots:



SUMMARY OF PART 5

- * FT protocol for achieving the HL in phase estimation under bit-flip noise
- * Robust state preparation: Repeated Stabilizer Measurements $\Rightarrow$ Threshold
- * Robust measurement: Repeated Non-destructive Measurements
- * S gate manipulation